

**NVIDIA Regional R&D and Production Hub
Network Testing Project Report**

**Cisco Packet Tracer Implementation
Professional Project Report**

Executive Summary

This report documents the design, implementation and validation of a secure enterprise network developed in Cisco Packet Tracer for the NVIDIA Regional R&D and Production Hub. The project implements a Spine-Leaf architecture with VLAN segmentation, DMZ services, ASA firewall protection, AAA authentication, RADIUS, Syslog, NAT, ACL enforcement, STP security and endpoint protection. Validation confirmed that the implemented controls operate as expected within the capabilities of Cisco Packet Tracer.

Project Objectives

- Design a scalable Spine-Leaf enterprise network.
- Provide secure communication between departments.
- Protect internal resources using VLANs, ACLs and an ASA firewall.
- Implement centralized authentication and logging.
- Validate all implemented security controls.

Network Overview

Core Components:

- Spine switches
- Leaf switches
- Cisco ASA Firewall
- Edge Router
- RADIUS, Syslog, DHCP, DNS and FTP Servers
- DMZ Web Server
- Department VLANs
- Internet simulation using ISP loopback.

Implemented Security Controls

Control	Purpose	Status
VLAN Segmentation	Separate departments	Implemented
ACLs	Restrict unauthorized access	Validated
ASA Firewall	Perimeter protection	Validated
NAT/PAT	Internet access	Validated
DMZ	Public services isolation	Validated
RADIUS AAA	Central authentication	Validated
Syslog	Central logging	Validated
SSH	Secure management	Validated
STP Security	Prevent topology attacks	Validated
Endpoint Firewall	Host protection	Validated

Validation Summary

Test	Result
Internet Connectivity	PASS
Guest VLAN	PASS
ACL Validation	PASS
DMZ Security	PASS
ASA Firewall	PASS
NAT	PASS
FTP Authentication	PASS
RADIUS	PASS
Syslog	PASS
Edge Router	PASS
STP Security	PASS
Endpoint Protection	PASS
Web Server	PASS

Packet Tracer Limitations

Some enterprise features are only partially simulated. ACL bindings may not always appear in interface output after reopening a project. Advanced firewall inspection, real Internet connectivity and full endpoint security capabilities are beyond the scope of Packet Tracer.

Conclusion

The project successfully demonstrates a secure enterprise network using Cisco technologies. Testing verified that routing, switching, VLAN isolation, firewall policies, NAT, AAA, logging and infrastructure hardening function correctly. The solution provides a strong foundation for a production deployment while acknowledging the limitations of the simulator.

Recommendations

- Deploy enterprise hardware for production.
- Enable advanced monitoring and SIEM integration.
- Implement DHCP Snooping and Dynamic ARP Inspection where supported.
- Perform periodic security audits and penetration testing.
- Maintain regular configuration backups.

Web Landing Page Test Report

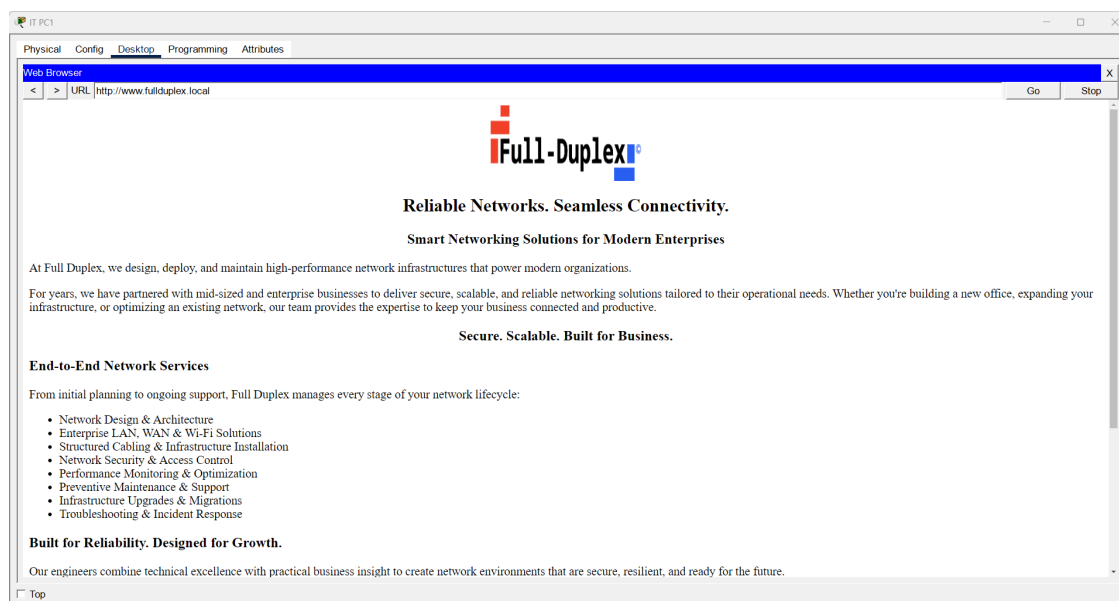
Objective

To verify that the internal web server is accessible through the network and that the company landing page loads correctly using the configured DNS name.

Test Procedure

1. Open IT PC1.
2. Navigate to Desktop > Web Browser.
3. Browse to <http://www.full duplex.local>.
4. Verify that the landing page loads successfully.

Evidence



Observation

The landing page loaded successfully using the domain name 'www.full duplex.local'. The company logo and website content were displayed correctly, confirming that DNS resolution, HTTP service and network connectivity are functioning as expected.

Packet Tracer Limitations

Cisco Packet Tracer hosts a simulated web server and does not provide access to a live website. The landing page is a basic HTML page intended to demonstrate DNS resolution, HTTP communication and network connectivity. Advanced web features such as HTTPS certificates, PHP, databases, JavaScript frameworks and dynamic content are not fully supported.

Conclusion

The landing page validation test passed successfully. Users can access the simulated company website through the configured domain name, demonstrating that the DNS service, web server and network infrastructure are operating correctly within the Cisco Packet Tracer environment.

Syslog Logging Test Report

Objective

This report summarizes the switch logging activity, analysis of SSH login attempts and subsequent syslog messaging in the Cisco Packet Tracer environment. The study focuses on how different user authentication attempts are logged by the system, including standard logins and fallback/unknown user scenarios.

Below screenshot- Console Activity: The CLI output confirms that a user logged in, entered privileged EXEC mode (en), and briefly entered global configuration mode (conf t) before exiting.

```
Leaf-1
Physical Config CLI Attributes pt-sm://net.netacad.cisco.activitysequencer/instructions.html
IOS Command Line Interface

Model revision number      : N0
Motherboard revision number : A0
Model number               : WS-C3650-24PS
System serial number       : FDO2031Q0TD

Press RETURN to get started!

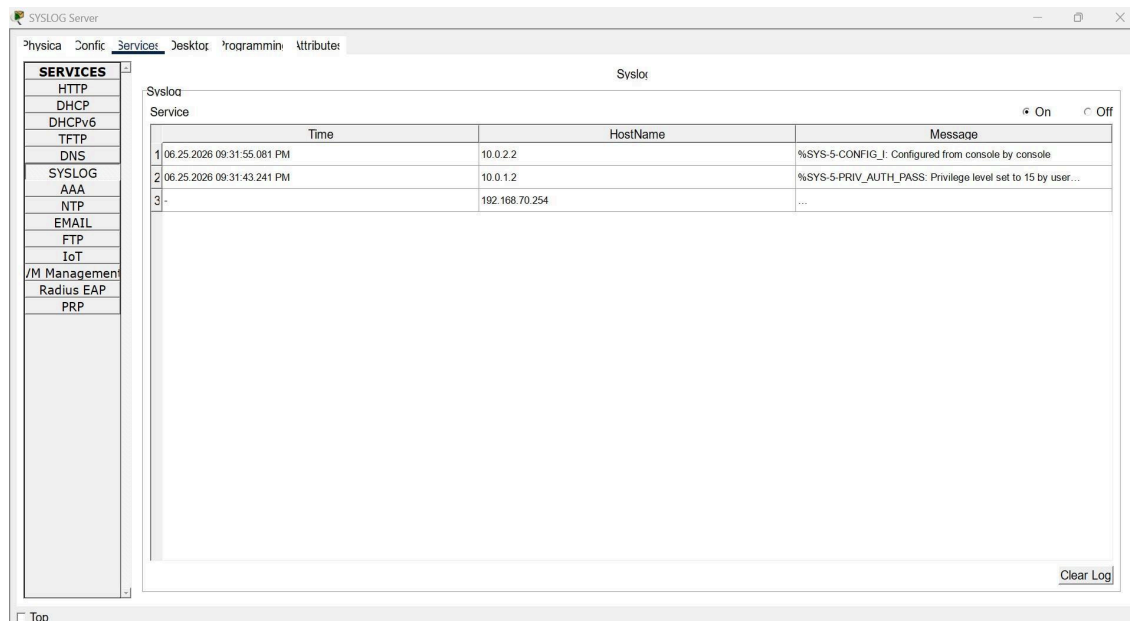
*****
* WARNING: Unauthorized access prohibited *
*****

User Access Verification

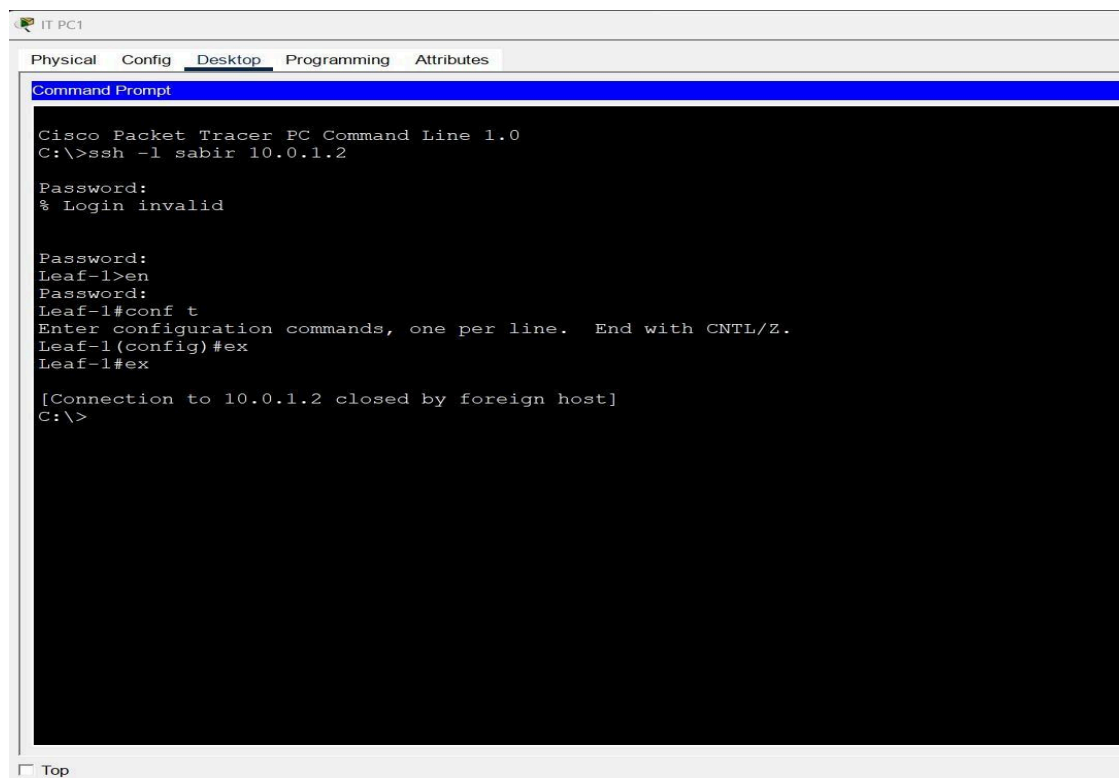
Username: user
Password:
Leaf-1>en
Password:
Leaf-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Leaf-1(config)#ex
Leaf-1#ex
```

Below screenshot- Syslog Captures: The Syslog server successfully logged the following events:

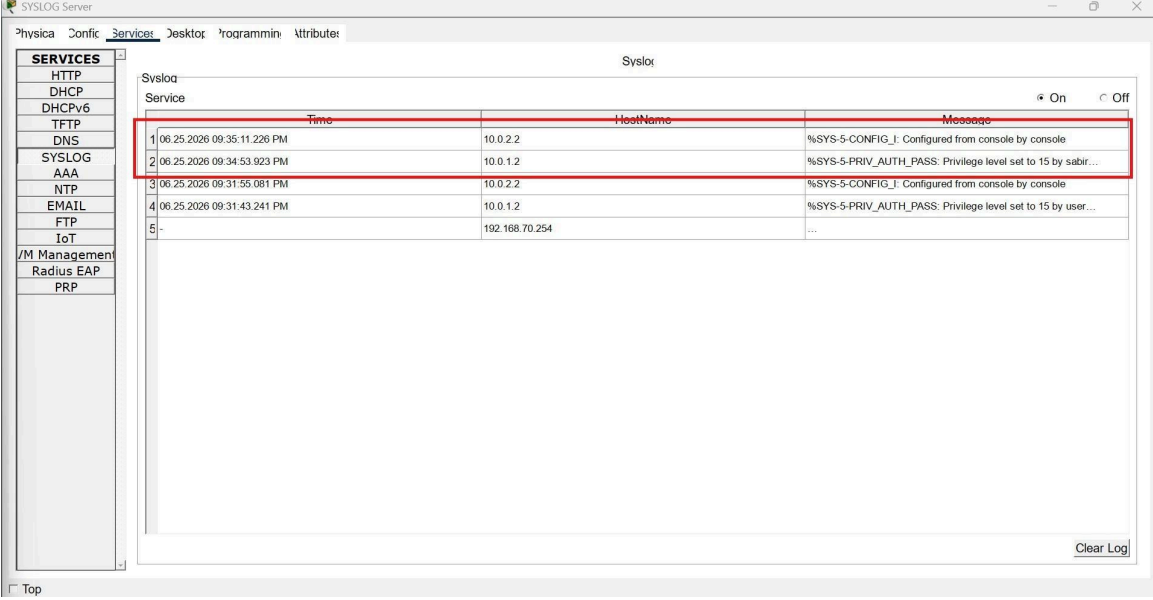
- Privilege Elevation: %SYS-5-PRIV_AUTH_PASS confirms the privilege level was set to 15 (administrative access).
- Configuration Changes: %SYS-5-CONFIG_I indicates that configuration commands were executed from the console.



Below screenshot- This confirms that a SSH connection is made has entered that a user (sabr) logged in, entered privileged EXEC mode (en), and briefly entered global configuration mode (conf t) before exiting.



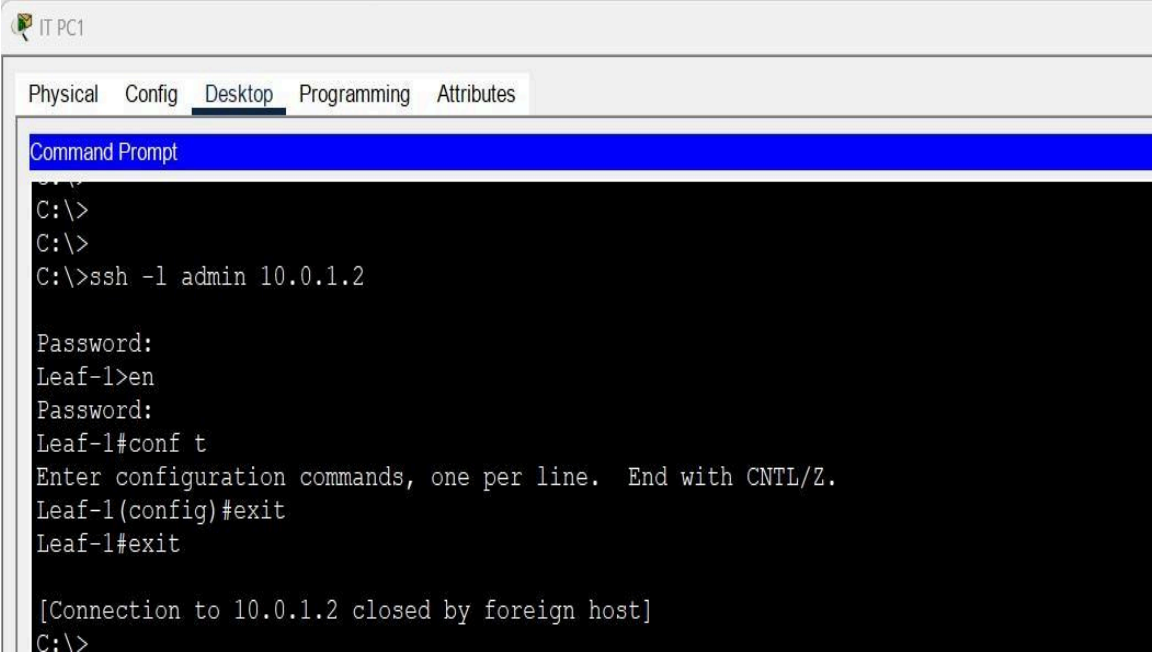
Below screenshot- %SYS-5-PRIV_AUTH_PASS: This message indicates that a user has successfully authenticated and elevated their privileges to level 15 (administrative/privileged EXEC mode). The log specifically names the user "sabir," showing that the system correctly identifies and tracks individual user identities when they log in.



The screenshot shows the Syslog Server interface with a list of log messages. A red box highlights the following messages:

	Time	HostName	Message
1	06/25/2026 09:35:11.226 PM	10.0.2.2	%SYS-5-CONFIG_I: Configured from console by console
2	06/25/2026 09:34:53.923 PM	10.0.1.2	%SYS-5-PRIV_AUTH_PASS: Privilege level set to 15 by sabir...
3	06/25/2026 09:31:55.081 PM	10.0.2.2	%SYS-5-CONFIG_I: Configured from console by console

Below screenshot- This confirms that AAA fallback to the local account has entered that a user logged in, entered privileged EXEC mode (en), and briefly entered global configuration mode (conf t) before exiting.



```

C:\>
C:\>
C:\>ssh -l admin 10.0.1.2

Password:
Leaf-1>en
Password:
Leaf-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Leaf-1(config)#exit
Leaf-1#exit

[Connection to 10.0.1.2 closed by foreign host]
C:\>
  
```

Below screenshot- Shows that the AAA fallback to the local account is working correctly. All expected events were successfully recorded on the Syslog server, confirming that both **AAA fallback authentication** and **centralized Syslog logging** are functioning correctly.

The screenshot shows the Syslog Server application window. On the left is a sidebar with a 'SERVICES' list including HTTP, DHCP, DHCPv6, TFTP, DNS, SYSLOG, AAA, NTP, EMAIL, FTP, IoT, /M Management, Radius EAP, and PRP. The main area is titled 'Syslog' and contains a table with columns 'Time', 'HostName', and 'Message'. The table has 7 rows of data. Rows 1 and 2 are highlighted with a red border. A 'Clear Log' button is at the bottom right of the table area.

	Time	HostName	Message
1	06.25.2026 09:43:09.501 PM	10.0.2.2	%SYS-5-CONFIG_I: Configured from console by console
2	06.25.2026 09:42:27.557 PM	10.0.1.2	%SYS-5-PRIV_AUTH_PASS: Privilege level set to 15 by unknown...
3	06.25.2026 09:35:11.228 PM	10.0.2.2	%SYS-5-CONFIG_I: Configured from console by console
4	06.25.2026 09:34:53.923 PM	10.0.1.2	%SYS-5-PRIV_AUTH_PASS: Privilege level set to 15 by sabir...
5	06.25.2026 09:31:55.081 PM	10.0.2.2	%SYS-5-CONFIG_I: Configured from console by console
6	06.25.2026 09:31:43.241 PM	10.0.1.2	%SYS-5-PRIV_AUTH_PASS: Privilege level set to 15 by user...
7	-	192.168.70.254	...

- **Observation**

Invalid SSH credentials were rejected while valid RADIUS and local administrator credentials authenticated successfully. The Syslog server recorded both authentication and configuration events with timestamps, confirming that logging and auditing are functioning correctly.

- **Conclusion**

All tests passed successfully.

- RADIUS authentication works correctly.
- Local administrator fallback works correctly.
- Unauthorized login attempts are rejected.
- Syslog records successful authentication and configuration events.

Spine Switch Interface Status Port Test Report

Objective: Verify the operational status of the Spine-1 interfaces and confirm routed links to the leaf switches.

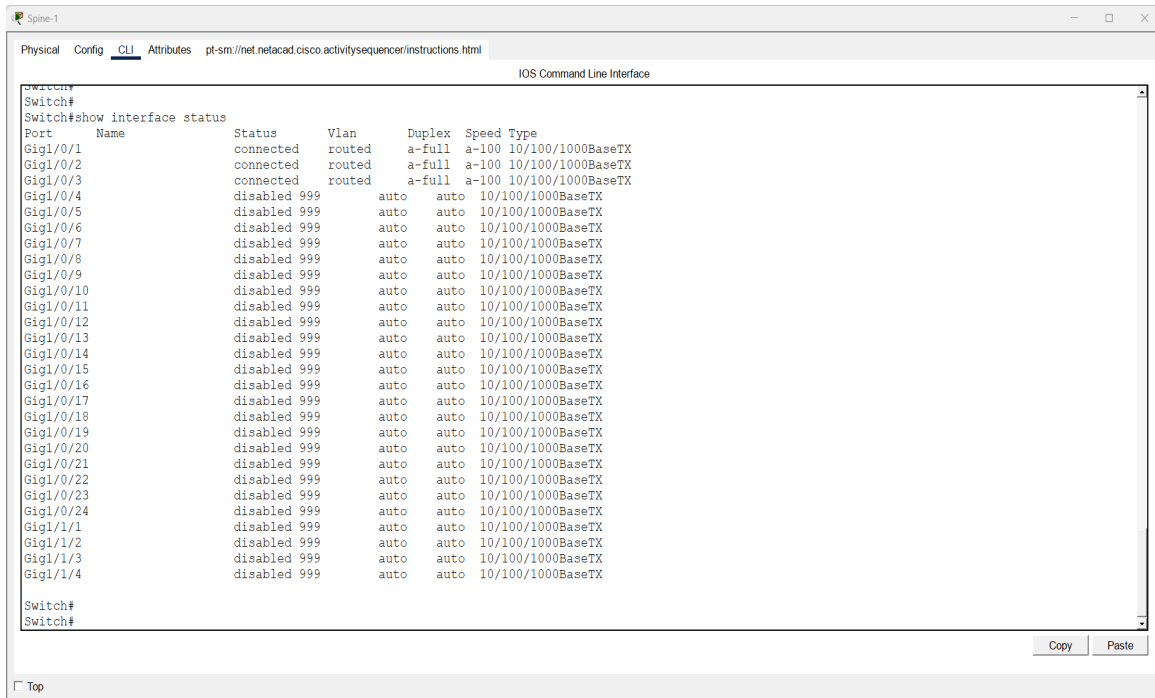
Test Result

The 'show interface status' command was executed on Spine-1. Interfaces Gig1/0/1, Gig1/0/2 and Gig1/0/3 are connected and configured as routed ports, confirming successful Layer 3 connectivity with the connected devices. All remaining interfaces are disabled and assigned to VLAN 999, reducing the risk of unauthorized access.

Observation

The Spine-1 switch is operating correctly. Active routed links are available for network forwarding while all unused interfaces have been securely disabled. This configuration supports a secure and reliable Spine-Leaf architecture.

Screenshot Evidence



Leaf Switch Interface Status Port Test Report

Objective: Verify the operational status of Leaf-1 interfaces, routed uplinks, trunk links and unused ports.

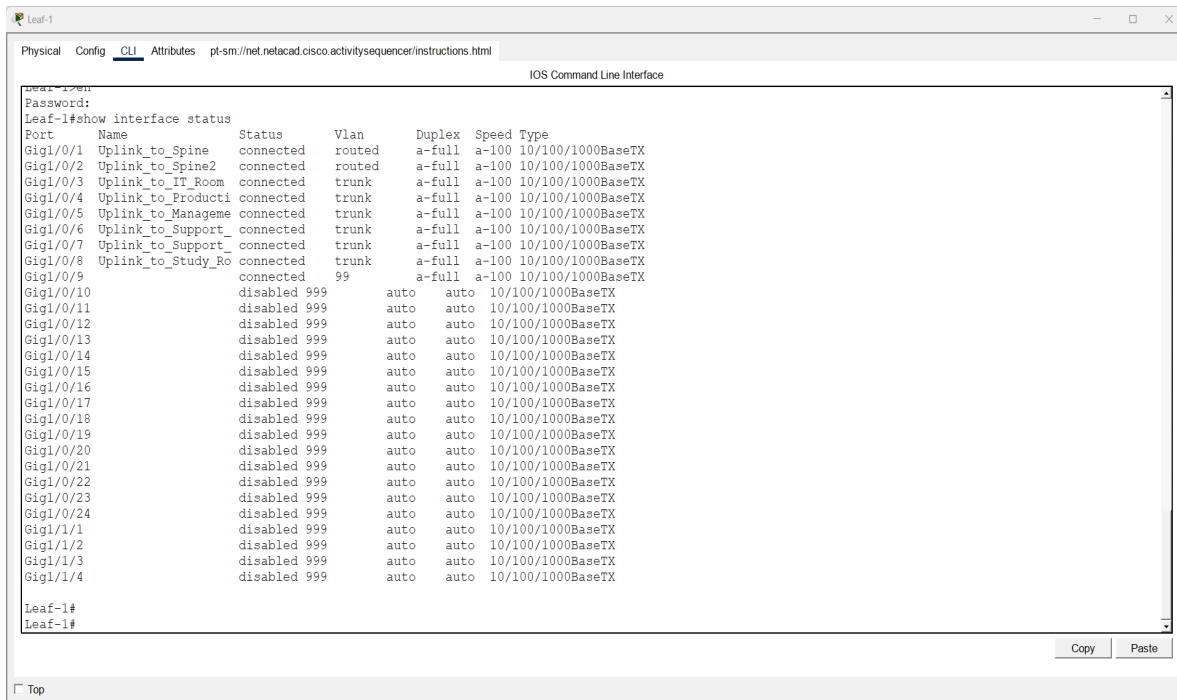
Test Result

The 'show interface status' command was executed on Leaf-1. Both routed uplinks to the Spine switches (Gig1/0/1 and Gig1/0/2) are connected and operational. Trunk links to the IT Room, Production, Management, Support and Study access switches are also connected, confirming successful Layer 2 connectivity. Unused interfaces have been disabled and assigned to VLAN 999 to reduce unauthorized access.

Observation

Leaf-1 is operating correctly. All required uplinks are active, trunk connections are established and unused ports are securely disabled. The switch is ready to support inter-VLAN communication and network traffic.

Screenshot Evidence



Access Switch Port Testing Report

Objective: Verify the operational status and VLAN assignment of the access switch interfaces.

Test Result

The 'show interfaces status' command was executed on the IT Access Switch. Ports Fa0/1 through Fa0/5 are connected and assigned to VLAN 10, confirming that the connected end devices are operational. GigabitEthernet0/1 is active and operating as a trunk link, allowing VLAN traffic to pass to the upstream switch. All unused access ports are disabled and placed in VLAN 999 to improve network security.

Observation

The switch is operating as expected. Active user ports are available in the correct VLAN, the trunk uplink is established successfully and unused interfaces have been secured by disabling them and assigning them to the unused VLAN 999.

Screenshot Evidence

IT-Access-Switch

Physical

Config

CLI

Attributes

IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Switch>en

Switch#show interfaces status

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1		connected	10	a-full	a-100	10/100BaseTX
Fa0/2		connected	10	a-full	a-100	10/100BaseTX
Fa0/3		connected	10	a-full	a-100	10/100BaseTX
Fa0/4		connected	10	a-full	a-100	10/100BaseTX
Fa0/5		connected	10	a-full	a-100	10/100BaseTX
Fa0/6		disabled	999	auto	auto	10/100BaseTX
Fa0/7		disabled	999	auto	auto	10/100BaseTX
Fa0/8		disabled	999	auto	auto	10/100BaseTX
Fa0/9		disabled	999	auto	auto	10/100BaseTX
Fa0/10		disabled	999	auto	auto	10/100BaseTX
Fa0/11		disabled	999	auto	auto	10/100BaseTX
Fa0/12		disabled	999	auto	auto	10/100BaseTX
Fa0/13		disabled	999	auto	auto	10/100BaseTX
Fa0/14		disabled	999	auto	auto	10/100BaseTX
Fa0/15		disabled	999	auto	auto	10/100BaseTX
Fa0/16		disabled	999	auto	auto	10/100BaseTX
Fa0/17		disabled	999	auto	auto	10/100BaseTX
Fa0/18		disabled	999	auto	auto	10/100BaseTX
Fa0/19		disabled	999	auto	auto	10/100BaseTX
Fa0/20		disabled	999	auto	auto	10/100BaseTX
Fa0/21		disabled	999	auto	auto	10/100BaseTX
Fa0/22		disabled	999	auto	auto	10/100BaseTX
Fa0/23		disabled	999	auto	auto	10/100BaseTX
Fa0/24		disabled	999	auto	auto	10/100BaseTX
Gig0/1		connected	trunk	a-full	a-100	10/100/1000BaseTX
Gig0/2		disabled	999	auto	auto	10/100/1000BaseTX

Switch#

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STP Security (BPDU Guard and Root Guard) Validation Report

Objective

Verify that BPDU Guard and Root Guard are correctly configured on the Access-Switches to protect the spanning tree topology from unauthorized devices and rogue root bridge elections.

Test Procedure

- Accessed the IT-Access-Switch using privileged EXEC mode.
- Executed the command: show spanning-tree summary.
- Executed the command: show running-config | include guard.
- Reviewed the output to verify BPDU Guard and Root Guard configuration.

Verification Screenshots

Figure 1. STP summary showing BPDU Guard status.

IT-Access-Switch

Physical

Config

CLI

Attributes

IOS Command Line Interface

Press RETURN to get started!

WARNING: Authorized access only. Unauthorized use is strictly prohibited.

User Access Verification

Password:

IT-access-switch>en

Password:

IT-access-switch#show spanning-tree summary

Switch is in pvst mode

Root bridge for: default IT

Extended system ID is enabled

Portfast Default is disabled

PortFast BPDU Guard Default is enabled

Portfast BPDU Filter Default is disabled

Loopguard Default is disabled

EtherChannel misconfig guard is disabled

UplinkFast is disabled

BackboneFast is disabled

Configured Pathcost method used is short

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	0	0
VLAN0010	0	0	0	5	5
3 vlans	0	0	0	5	5

IT-access-switch#

☐ Top

Figure 2. Running configuration confirming BPDU Guard and Root Guard.

[illegible]

Results

Security Feature	Status	Evidence
BPDU Guard	Passed	PortFast BPDU Guard Default is enabled and interface configuration includes "spanning-tree bpduguard enable".
Root Guard	Passed	Running configuration includes "spanning-tree guard root".

Observation

The switch configuration confirms that BPDU Guard and Root Guard are enabled. BPDU Guard protects access ports from receiving unexpected BPDU messages while Root Guard prevents downstream switches from becoming the root bridge. The STP summary shows normal operation with no blocked or inconsistent ports.

Conclusion

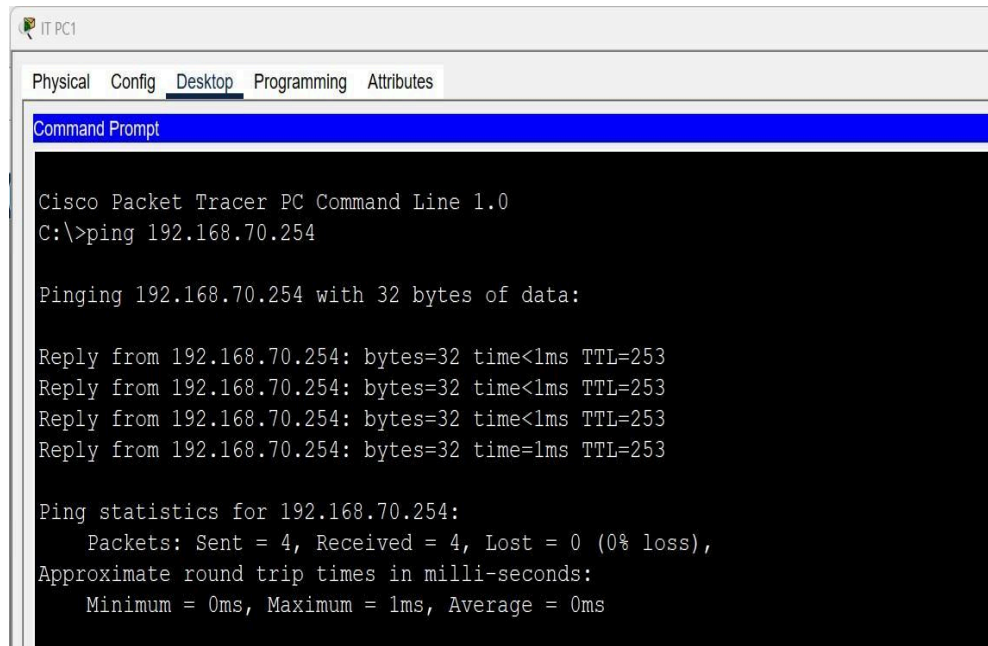
The STP security validation was successful. Both BPDU Guard and Root Guard are configured correctly and provide additional protection against spanning tree attacks and accidental topology changes.

RADIUS Server Test Report

Objective

Verify RADIUS authentication, SSH access, local fallback authentication and RADIUS accounting log.

Figure 1 - RADIUS Connectivity Test



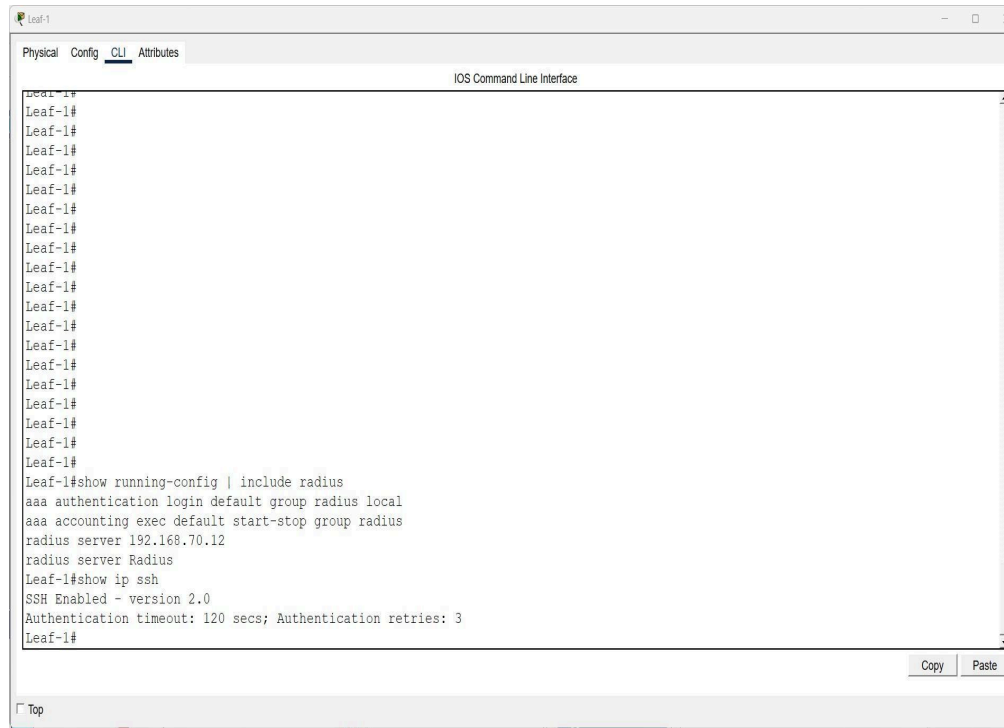
Ping to the RADIUS server was successful, confirming connectivity.

Figure 2 - RADIUS Configuration



Verified the RADIUS server configuration on Leaf-1.

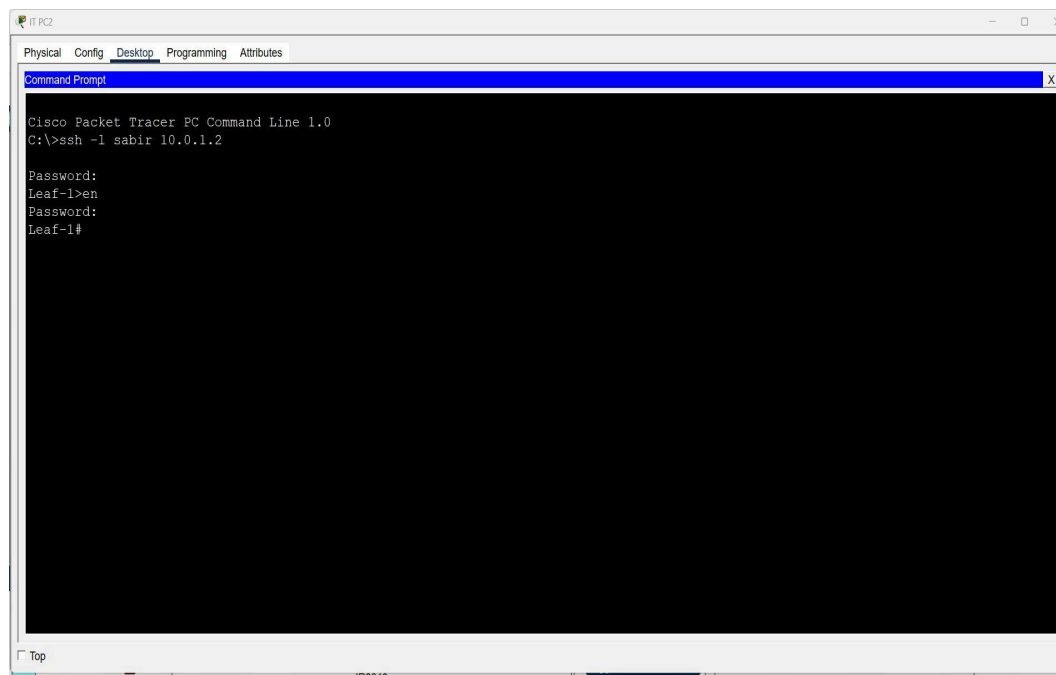
Figure 3 - AAA and SSH Configuration

A screenshot of a Cisco Packet Tracer interface showing the CLI of a device named 'Leaf-1'. The 'CLI' tab is selected. The command history shows multiple 'Leaf-1#' prompts. The configuration includes 'aaa authentication login default group radius local', 'aaa accounting exec default start-stop group radius', and 'radius server 192.168.70.12 radius server Radius'. The 'show ip ssh' command output indicates 'SSH Enabled - version 2.0' and 'Authentication timeout: 120 secs; Authentication retries: 3'.

```
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#
Leaf-1#show running-config | include radius
aaa authentication login default group radius local
aaa accounting exec default start-stop group radius
radius server 192.168.70.12
radius server Radius
Leaf-1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
Leaf-1#
```

Leaf-1 is configured to use RADIUS with local fallback and SSH v2.

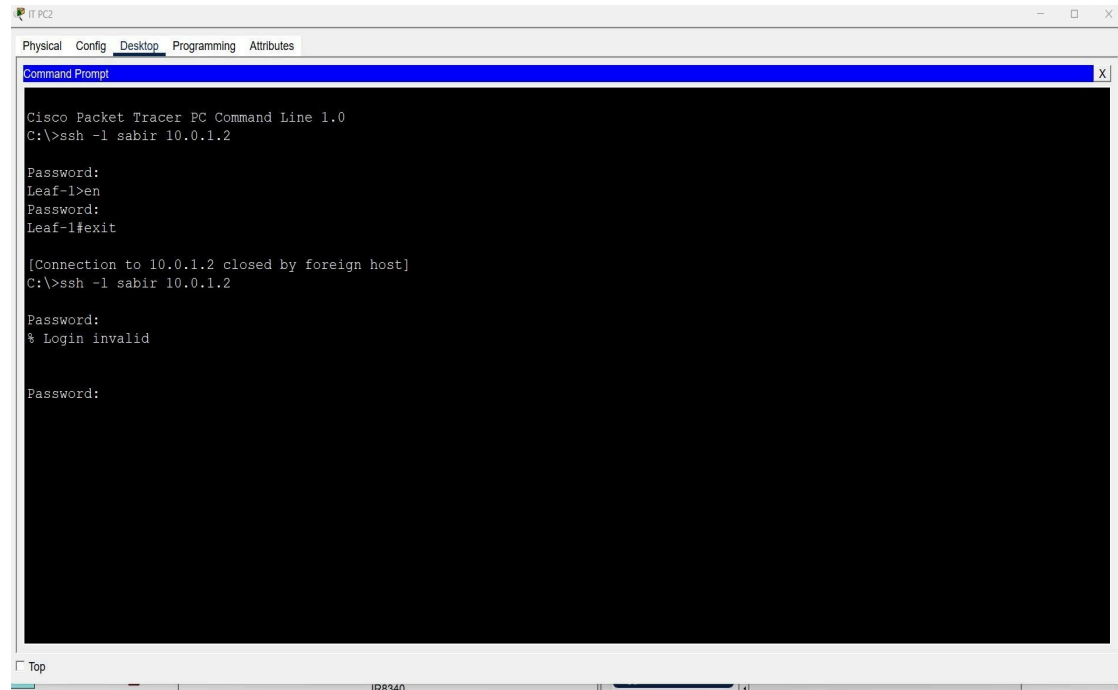
Figure 4 - Successful SSH Login (RADIUS User)

A screenshot of a Cisco Packet Tracer interface showing the Command Prompt of a device named 'PC2'. The 'Command Prompt' tab is selected. The command history shows the execution of 'ssh -l sabir 10.0.1.2', followed by 'Password:', 'Leaf-1>en', 'Password:', and 'Leaf-1#'.

```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l sabir 10.0.1.2
Password:
Leaf-1>en
Password:
Leaf-1#
```

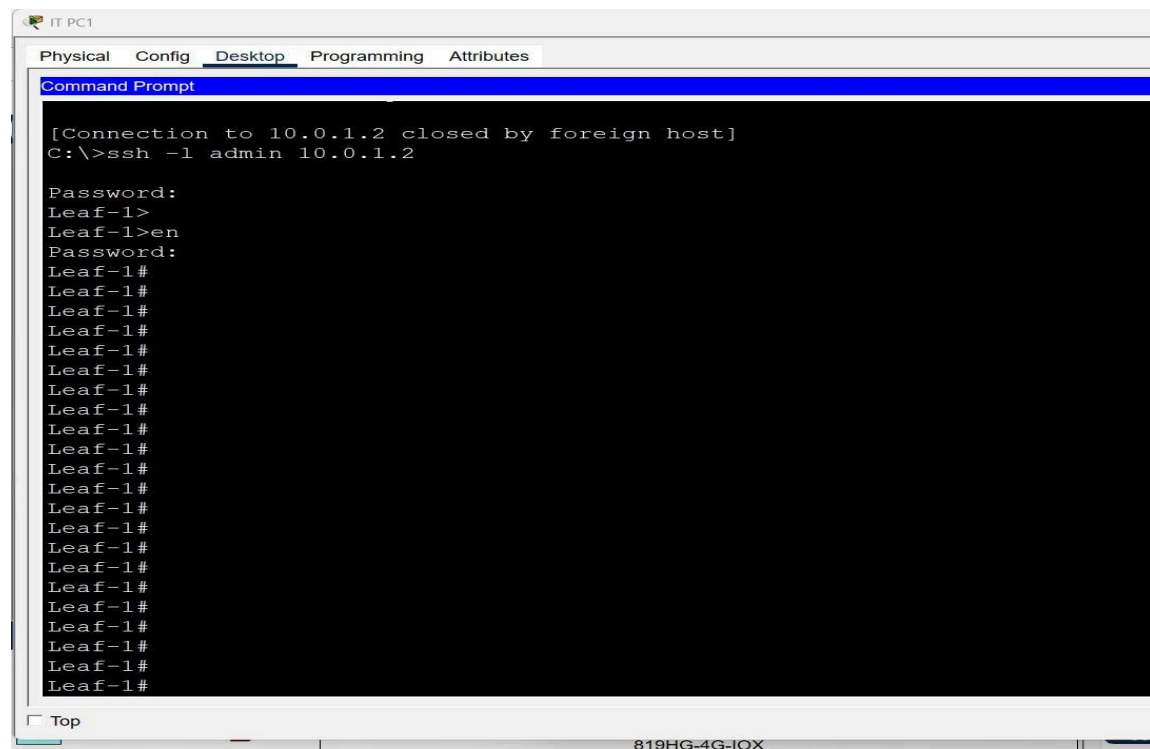
SSH login using user 'sabir' was successful through the RADIUS server.

Figure 5 - Wrong Password Test



Authentication failed when an incorrect password was entered.

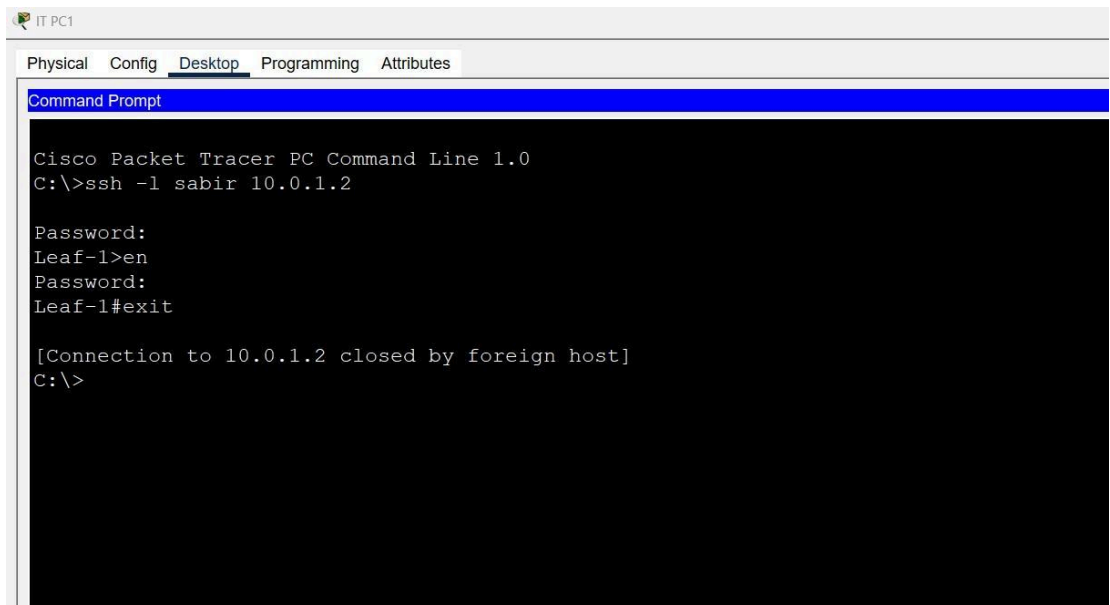
Figure 6 - Local Fallback Login



After disabling the RADIUS server, the local admin account logged in successfully.

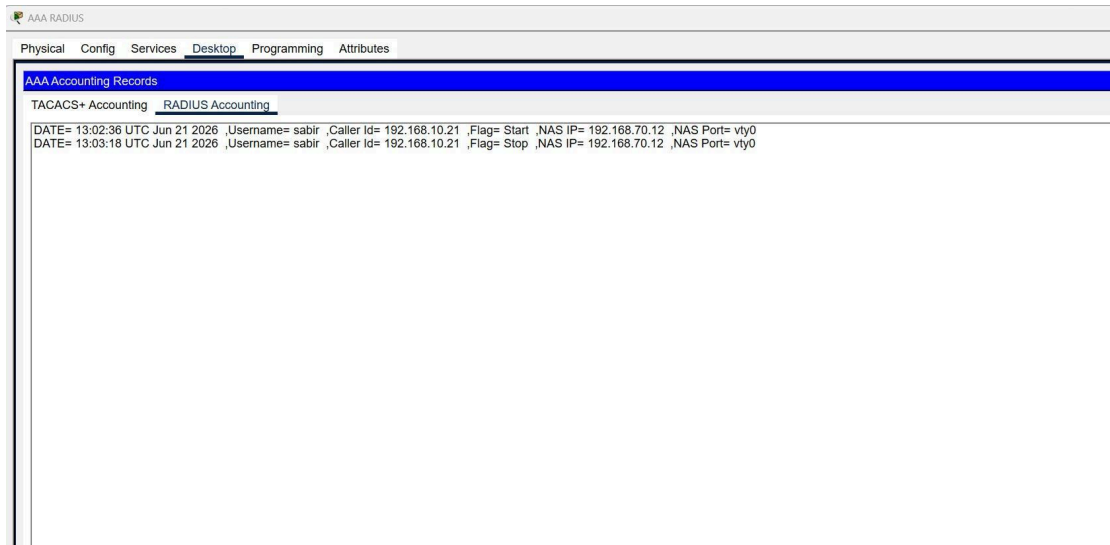
RADIUS Accounting Test Report

Figure 7 – Login via SSH User 'sabir'



Successful SSH login authenticated through the RADIUS server.

Figure 8 - RADIUS Accounting Log



RADIUS accounting records showing the Start and Stop events for user "sabir".

Conclusion

Testing confirmed that RADIUS authentication works correctly. Valid users authenticated successfully, invalid credentials were rejected and the local administrator account provided successful fallback access when the RADIUS server was unavailable. The accounting log successfully recorded both the Start and Stop events, including the username, caller IP address, NAS IP address and VTY port. This confirms that RADIUS accounting is functioning correctly.

Printer Connectivity Test Report

Test Item: Management Printer Connectivity

Objective

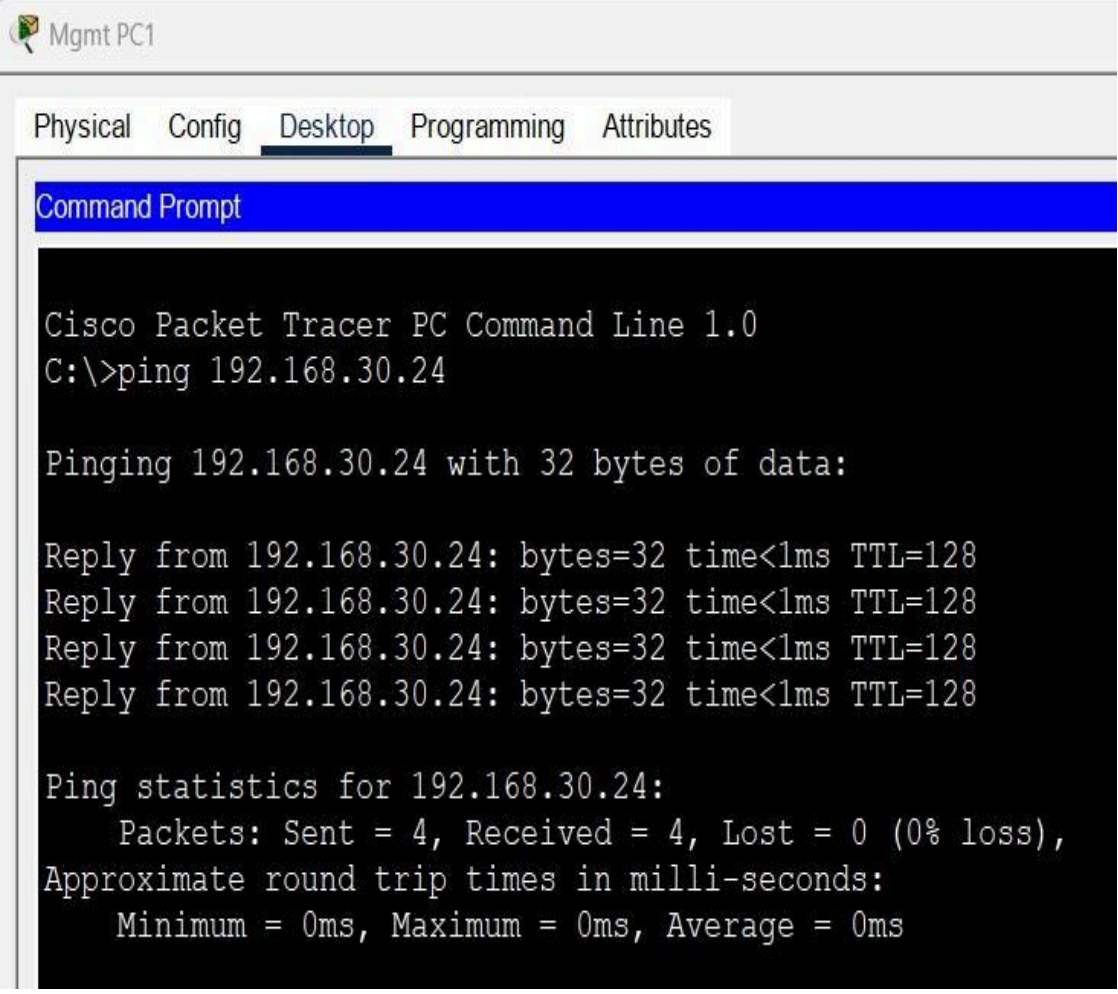
To verify that the management printer is correctly configured with a static IP address and is reachable from a management workstation within VLAN 30.

Test Environment

Parameter	Value
Printer	Mgmt Printer 1
IP Address	192.168.30.24/24
Gateway	192.168.30.254
Source Device	Management PC1
Test	ICMP Ping

Evidence

Figure 1. Successful ICMP ping from Management PC1 to the management printer.



```
Mgmt PC1
Physical Config Desktop Programming Attributes
Command Prompt

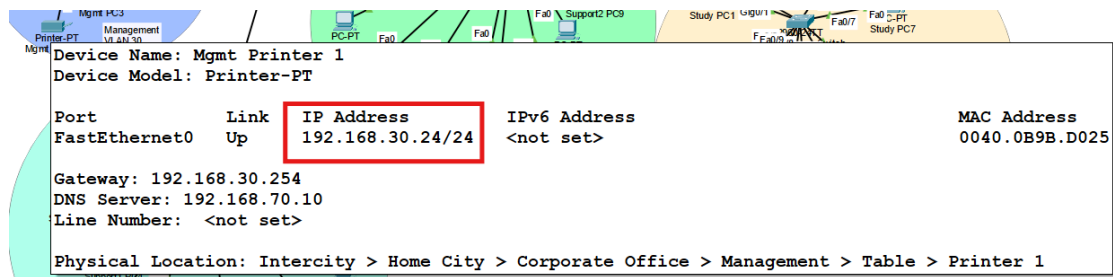
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.24

Pinging 192.168.30.24 with 32 bytes of data:

Reply from 192.168.30.24: bytes=32 time<1ms TTL=128
Reply from 192.168.30.24: bytes=32 time<1ms TTL=128
Reply from 192.168.30.24: bytes=32 time<1ms TTL=128
Reply from 192.168.30.24: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.30.24:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 2. Printer network configuration showing the configured static IP address.



The image shows a network configuration window for a device named 'Mgmt Printer 1'. The window displays various configuration details, including the device model, port status, IP address, IPv6 address, MAC address, gateway, DNS server, and physical location. The IP address '192.168.30.24/24' is highlighted with a red box.

Port	Link	IP Address	IPv6 Address	MAC Address
FastEthernet0	Up	192.168.30.24/24	<not set>	0040.0B9B.D025

Gateway: 192.168.30.254
DNS Server: 192.168.70.10
Line Number: <not set>

Physical Location: Intercity > Home City > Corporate Office > Management > Table > Printer 1

Test Procedure

- Open Command Prompt on Management PC1.
- Run: ping 192.168.30.24
- Verify that all ICMP replies are received.
- Confirm the printer IP configuration matches the assigned address.

Observation

All four ICMP echo requests received successful replies with 0% packet loss. The printer is configured with the static IP address 192.168.30.24/24 and the default gateway 192.168.30.254. This confirms correct Layer 3 connectivity between the management workstation and the network printer.

Result

Status: PASS

Conclusion

The management printer is correctly addressed and accessible from the Management VLAN. This confirms that printer connectivity and basic network communication are functioning as expected.

NAT Test Report

Test Objective

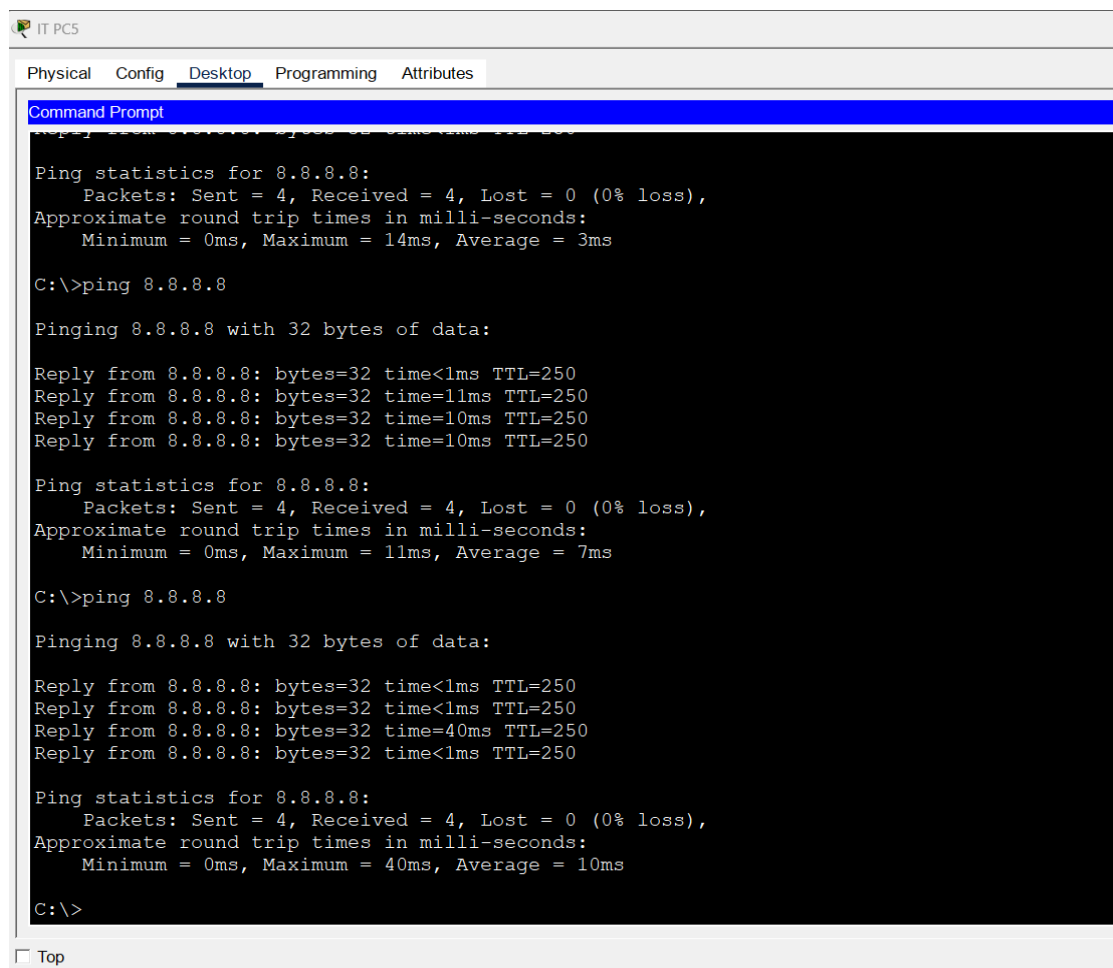
Verify that the ASA Firewall performs NAT/PAT for internal users and static NAT for DMZ services.

Procedure

1. Open Workstation EX-IT PC.
2. Execute:

ping 8.8.8.8

3. Verify successful replies.
4. On ASA execute:



The screenshot shows a Windows Command Prompt window titled "IT PCs" with tabs for Physical, Config, Desktop, Programming, and Attributes. The Command Prompt displays the output of a ping command to 8.8.8.8. The first ping shows statistics for 4 packets sent and received with 0% loss, and round trip times ranging from 0ms to 14ms. The second ping shows individual replies with times of <1ms, 11ms, 10ms, and 10ms. The third ping shows statistics for 4 packets sent and received with 0% loss, and round trip times ranging from 0ms to 40ms. The window has a "Top" button at the bottom left.

```
IT PCs
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 8.8.8.8: bytes=32 time=11ms TTL=250

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=250
Reply from 8.8.8.8: bytes=32 time=11ms TTL=250
Reply from 8.8.8.8: bytes=32 time=10ms TTL=250
Reply from 8.8.8.8: bytes=32 time=10ms TTL=250

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 7ms

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=250
Reply from 8.8.8.8: bytes=32 time<1ms TTL=250
Reply from 8.8.8.8: bytes=32 time=40ms TTL=250
Reply from 8.8.8.8: bytes=32 time<1ms TTL=250

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 40ms, Average = 10ms

C:\>
```

☐ Top

```
ASA Firewall
Physical Config CLI Attributes
IOS Command Line Interface

ciscoasa>show xlate
^
% Invalid input detected at '^' marker.

ciscoasa>en
Password:
ciscoasa#
ciscoasa#
ciscoasa#en
^
% Invalid input detected at '^' marker.

ciscoasa#
ciscoasa#
ciscoasa#
ciscoasa#show xlate
2 in use, 2 most used
Flags: D - DNS, e - extended, I - identity, i - dynamic, r - portmap, s - static, T - twice, N - net-to-net
ICMP PAT from inside:192.168.10.28/5 to outside:203.0.113.6/12670 flags i idle 00:00:26, timeout 0:00:30
NAT from dmz:192.168.80.10/32 to outside:203.0.113.10/32 flags s idle 00:32:07, timeout 0:00:00

ciscoasa#show nat detail
^
% Invalid input detected at '^' marker.

ciscoasa#show conn
^
% Invalid input detected at '^' marker.

ciscoasa#clear xlate
^
% Invalid input detected at '^' marker.

ciscoasa#
```

show xlate

Results

Source IP	Translated IP	Status
192.168.10.28	203.0.113.6	PASS
192.168.80.10	203.0.113.10	PASS

Conclusion

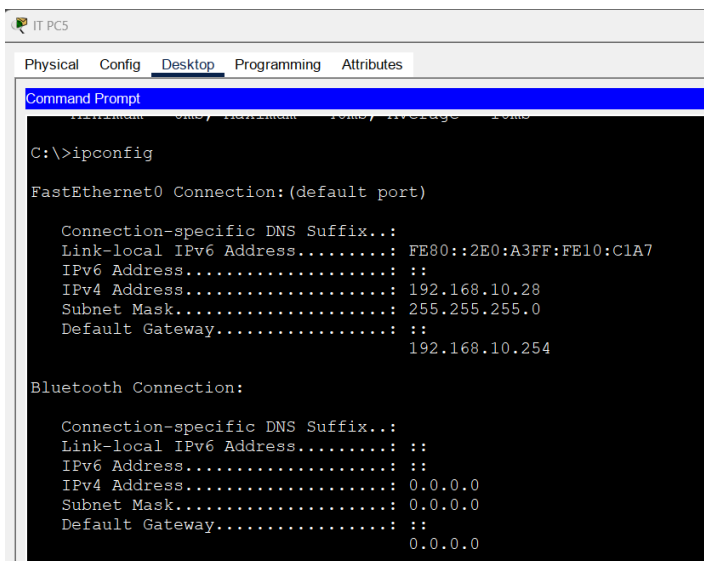
- Dynamic PAT for inside users is operational.
- Static NAT for DMZ web server is operational.
- Internet connectivity through the ASA firewall is functioning correctly.

IP Spoofing Test Report

Objective

To verify whether a workstation can gain unauthorized access to another VLAN by manually changing its IP address.

Original One



```
IT PCS
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig

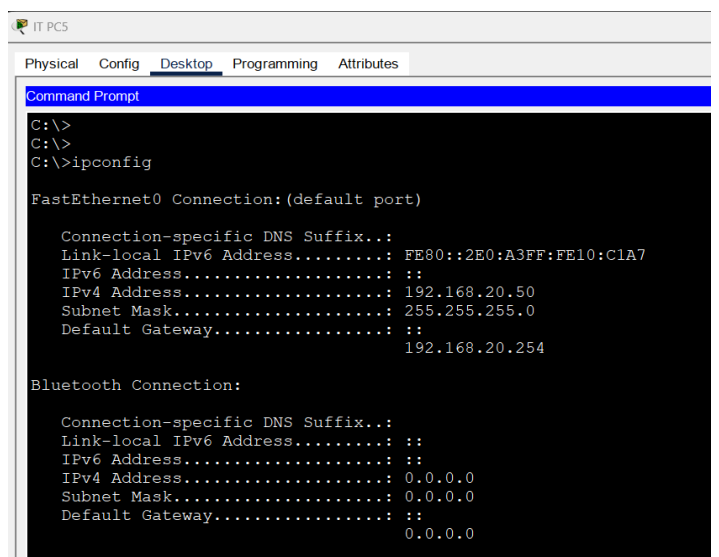
FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:A3FF:FE10:C1A7
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.10.28
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.10.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0
```

After IP Spoofing



```
IT PCS
Physical Config Desktop Programming Attributes
Command Prompt
C:\>
C:\>
C:\>ipconfig

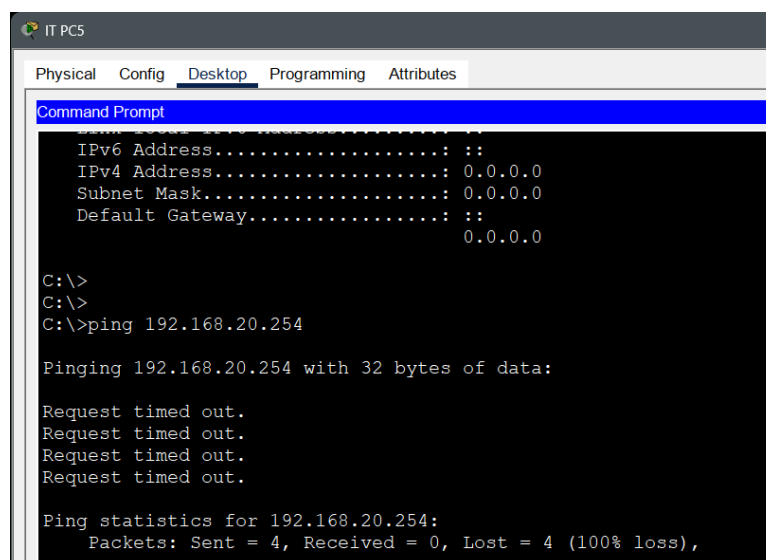
FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::2E0:A3FF:FE10:C1A7
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.20.50
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: 192.168.20.254

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0
```

Result



```
IT PCS
Physical Config Desktop Programming Attributes
Command Prompt
IPv6 Address . . . . .: ::
IPv4 Address . . . . .: 0.0.0.0
Subnet Mask . . . . .: 0.0.0.0
Default Gateway . . . . .: 0.0.0.0

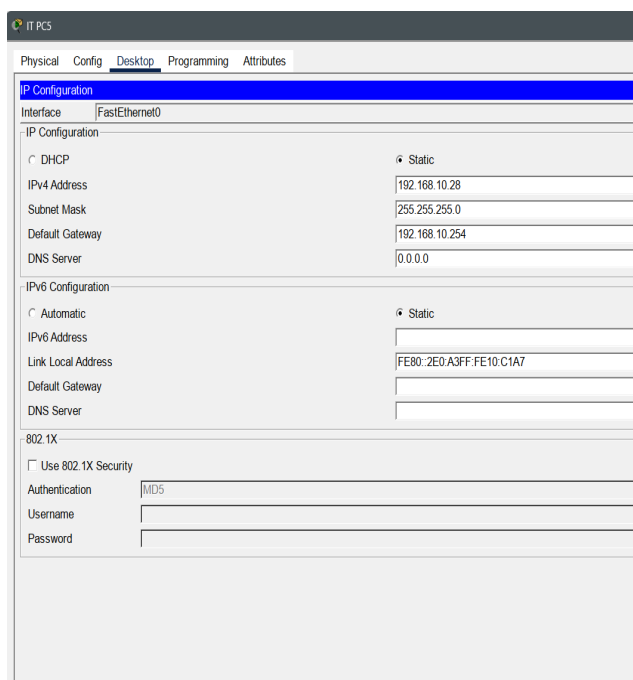
C:\>
C:\>
C:\>ping 192.168.20.254

Pinging 192.168.20.254 with 32 bytes of data:

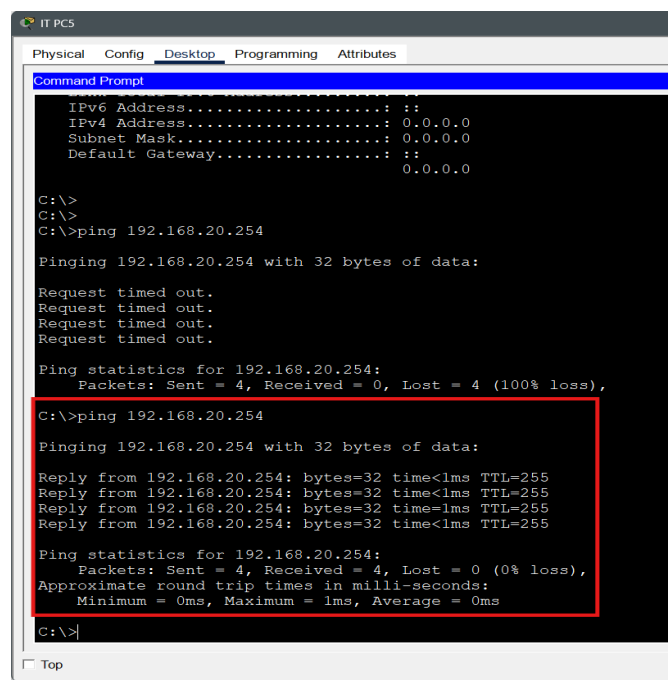
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.254:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Change it back Like before (Original)



After result



Test Performed

An IT workstation originally configured with IP address **192.168.10.28 (VLAN 10)** was manually changed to **192.168.20.50 (VLAN 20)** to simulate an IP spoofing attack.

The workstation then attempted to communicate with the Production VLAN gateway (**192.168.20.254**).

Results

- Initial ping attempts timed out during ARP resolution.

Security Analysis

The spoofing attempt showed that a user can manually change an IP address and be treated as a device from another subnet. While VLAN segmentation and ACL policies are functioning additional security controls are recommended to prevent IP spoofing.

Conclusion

The network successfully enforced VLAN segmentation and ACL-based access control.

Internet Connectivity Test Report

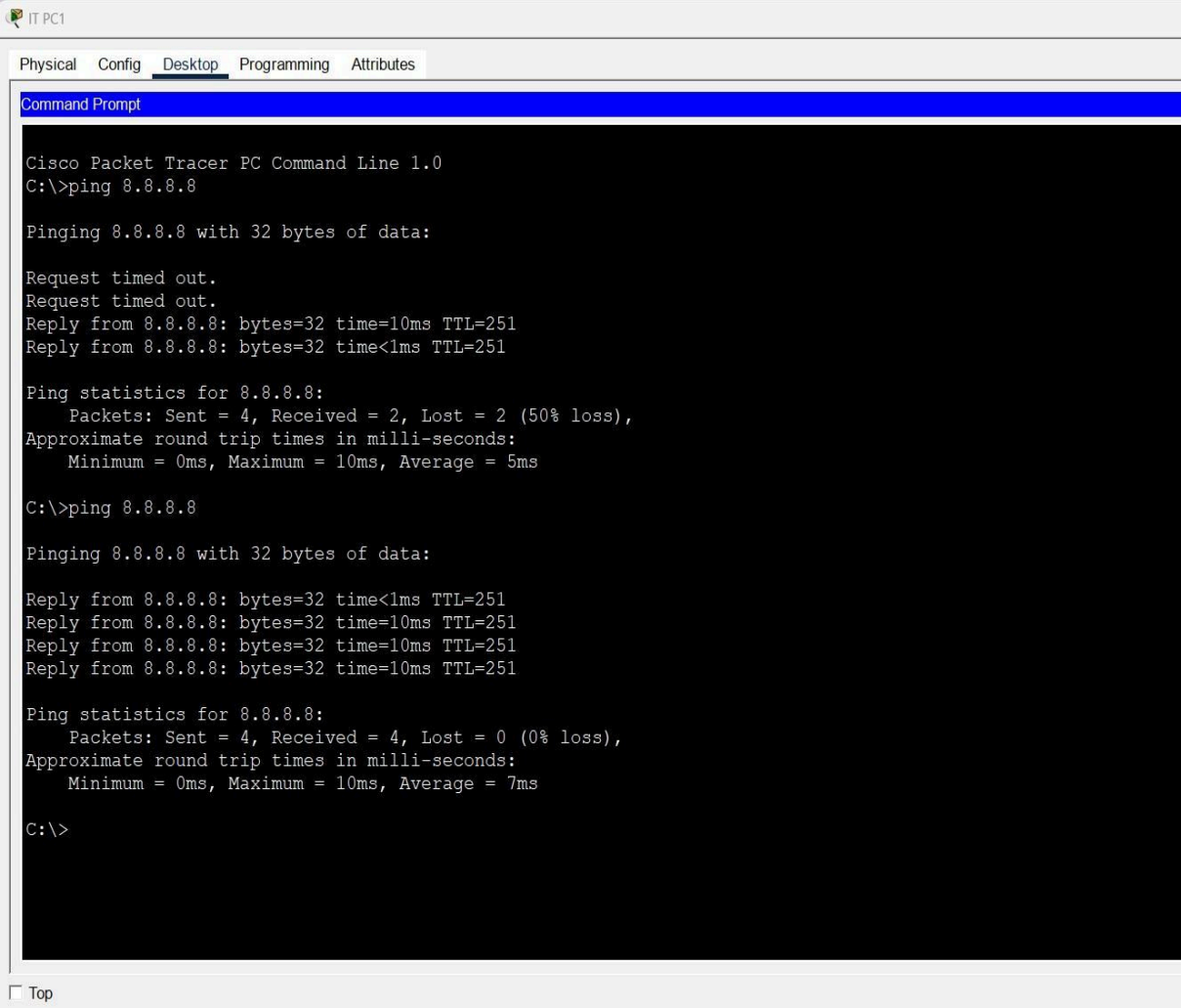
Objective

To verify that the internal network can successfully reach the simulated Internet in Cisco Packet Tracer.

Test Performed

From IT PC1, the command 'ping 8.8.8.8' was executed. The first ping experienced two timeouts, which is normal in Packet Tracer while ARP tables are populated. The second ping completed successfully with 0% packet loss, confirming end-to-end connectivity.

Evidence



```
IT PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 8.8.8.8: bytes=32 time=10ms TTL=251
Reply from 8.8.8.8: bytes=32 time<1ms TTL=251

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms

C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=251
Reply from 8.8.8.8: bytes=32 time=10ms TTL=251
Reply from 8.8.8.8: bytes=32 time=10ms TTL=251
Reply from 8.8.8.8: bytes=32 time=10ms TTL=251

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 7ms

C:\>
```

Observation

Successful ICMP replies from 8.8.8.8 confirm that the workstation can reach the simulated Internet through the network. The TTL value and response times indicate that packets were successfully routed to the destination.

Packet Tracer Limitations

Packet Tracer does not connect to the real Internet. In this project, the address 8.8.8.8 is represented by a loopback interface on the simulated ISP router. This loopback acts as a virtual Internet destination for testing routing, NAT and connectivity. Although the test proves the network configuration is working correctly inside the simulation, it does not verify connectivity to Google's real DNS server or the public Internet.

Conclusion

The Internet connectivity test passed successfully. The network can reach the simulated Internet destination, demonstrating correct routing and connectivity within the Cisco Packet Tracer environment.

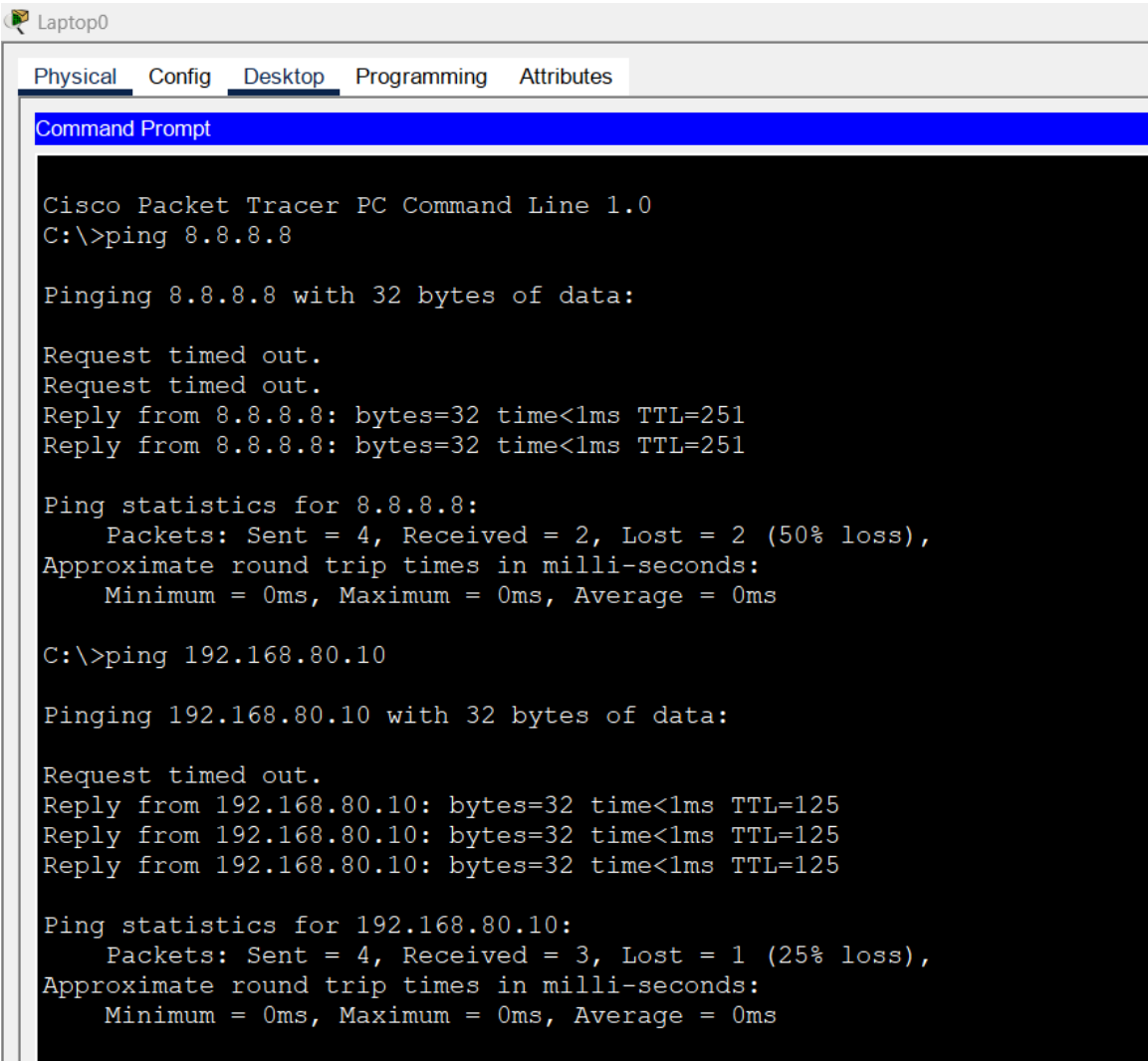
Guest VLAN Connectivity Test Report

Objective: Verify that Guest VLAN users can access the Internet and public website while being blocked from internal resources.

Test	Destination	Result	Status
Internet	8.8.8.8	Reachable	PASS
Public Website	192.168.80.10	Reachable	PASS
Internal Server	192.168.70.10	Blocked	PASS
Internal Department	192.168.10.21	Blocked	PASS

Evidence

Screenshot 1 - Internet and public website access



```
Laptop0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 8.8.8.8: bytes=32 time<1ms TTL=251
Reply from 8.8.8.8: bytes=32 time<1ms TTL=251

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.80.10

Pinging 192.168.80.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.80.10: bytes=32 time<1ms TTL=125
Reply from 192.168.80.10: bytes=32 time<1ms TTL=125
Reply from 192.168.80.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.80.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Screenshot 2 - Internal server and department access blocked

```
C:\>ping 192.168.70.10

Pinging 192.168.70.10 with 32 bytes of data:

Reply from 192.168.99.254: Destination host unreachable.
Reply from 192.168.99.254: Destination host unreachable.
Reply from 192.168.99.254: Destination host unreachable.
Reply from 192.168.99.254: Destination host unreachable.

Ping statistics for 192.168.70.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.21

Pinging 192.168.10.21 with 32 bytes of data:

Reply from 192.168.99.254: Destination host unreachable.
Reply from 192.168.99.254: Destination host unreachable.
Reply from 192.168.99.254: Destination host unreachable.
Reply from 192.168.99.254: Destination host unreachable.

Ping statistics for 192.168.10.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Observation

During testing it was observed that after reopening the Cisco Packet Tracer project, the Access Control List (ACL) remained configured but was no longer applied to the interface. The ip access-group binding had been removed, allowing traffic that should have been blocked to pass through. After reapplying the ACL to the correct interface, the expected security policy was restored and all connectivity tests produced the correct results.

Conclusion

The Guest VLAN security policy was successfully validated. Guests can access the Internet and the public web server but cannot reach internal servers or department devices. The ACL configuration is operating as expected.

FTP Server Test Report

Objective

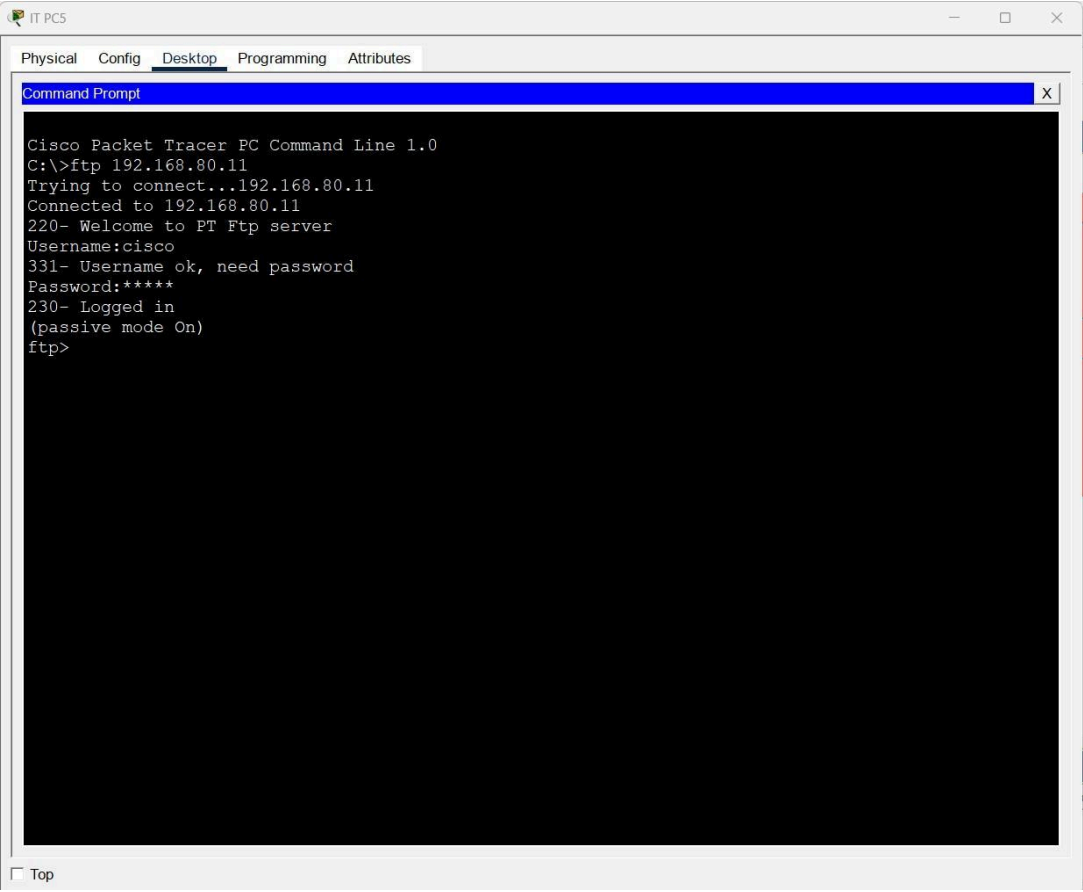
Verify that the FTP server requires valid user authentication and prevents unauthorized access.

Test Result

The FTP server was configured with dedicated user accounts and password-based authentication. A client workstation successfully connected to the FTP service using valid credentials. The server granted access only after successful authentication, confirming that anonymous access is disabled and user authentication is enforced.

Screenshots

Figure 1. Successful FTP login from the client workstation.

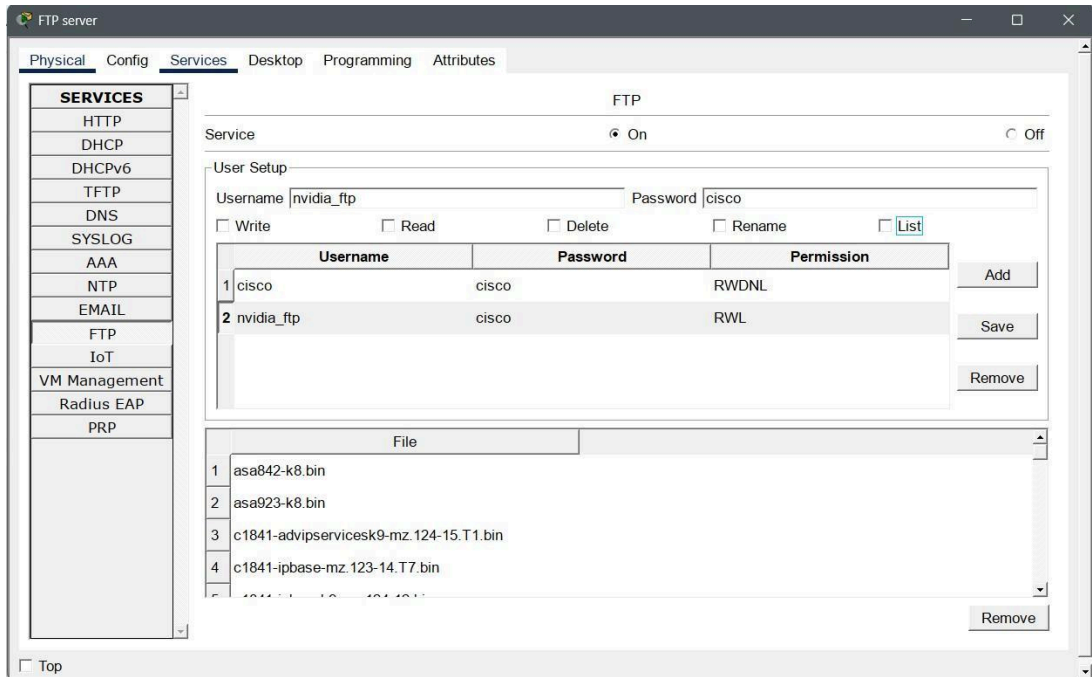


The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'IT PCs'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' selected. The command prompt shows the following sequence of commands and responses:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 192.168.80.11
Trying to connect...192.168.80.11
Connected to 192.168.80.11
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>
```

The window also includes a 'Top' button at the bottom left.

Figure 2. FTP server user configuration and permissions.



Observation

The login was completed successfully with the configured username and password. This confirms that secure account management is functioning correctly and only authorized users can access the FTP server.

Conclusion

The FTP server hardening test passed successfully. Authentication controls are working as expected and unauthorized access is prevented.

Endpoint Test Report

Purpose

This section evaluates the protection applied to end-user devices in the Packet Tracer network. The objective is to verify that workstations can defend themselves against unwanted connections by using local firewall policies. Although Packet Tracer does not provide enterprise endpoint security features, it allows the basic concept of host-based access control to be demonstrated.

Implemented Control

A local firewall policy was deployed on the IT workstation. The policy denies all IP traffic originating from the remote host 192.168.50.20. This configuration prevents the selected device from communicating with the protected workstation while leaving the remainder of the network unchanged.

Firewall Setting	Configured Value
Action	Deny
Protocol	IP
Source Address	192.168.50.20
Wildcard Mask	0.0.0.0
Destination Port	Any

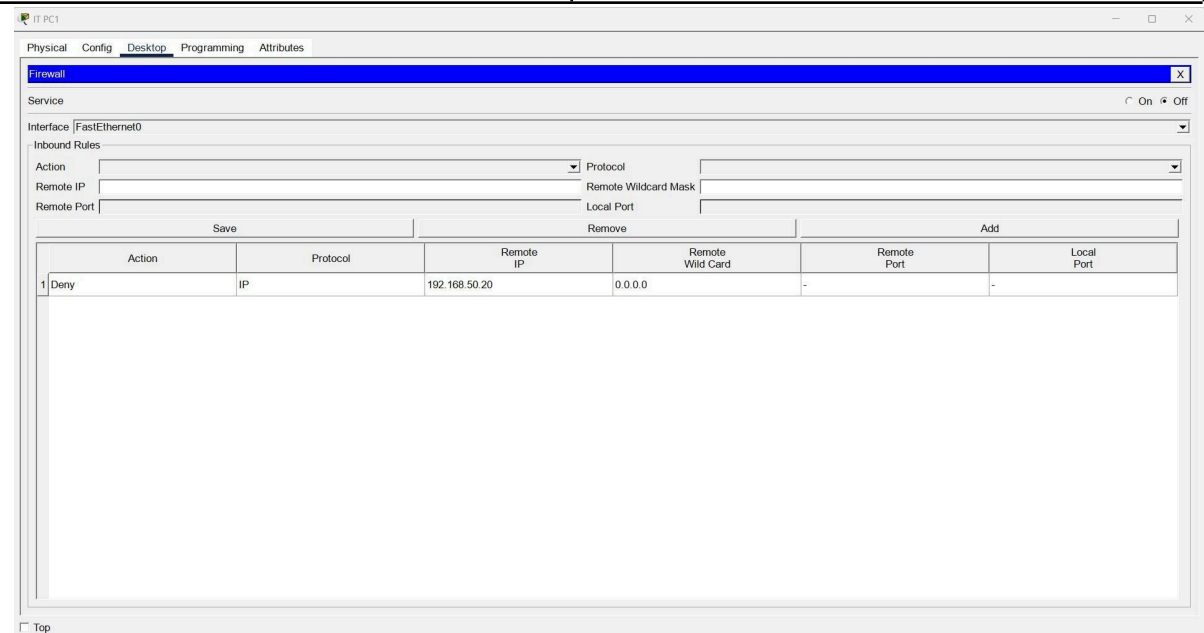


Figure 1. Local firewall policy configured on the protected workstation.

Verification Procedure

Connectivity tests were performed from the remote workstation before and after enabling the firewall. The first test confirmed normal communication. The second test confirmed that the firewall successfully blocked all ICMP traffic from the specified source.

The screenshot shows a Cisco Packet Tracer PC Command Line 1.0 window. The 'Desktop' tab is selected. The command prompt shows a successful ping to 192.168.10.21 before the firewall is activated, with 4 packets received and 0% loss. After activating the firewall, the ping fails with 0 packets received and 100% loss. Red boxes highlight the successful ping results and the 100% loss result. Text annotations 'Before activating the firewall from workstation.' and 'After activating the firewall from workstation.' are present.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.21

Pinging 192.168.10.21 with 32 bytes of data:

Reply from 192.168.10.21: bytes=32 time=10ms TTL=127
Reply from 192.168.10.21: bytes=32 time<1ms TTL=127
Reply from 192.168.10.21: bytes=32 time=10ms TTL=127
Reply from 192.168.10.21: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms

C:\>ping 192.168.10.21

Pinging 192.168.10.21 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 2. Successful communication before the firewall was enabled followed by complete packet loss after the rule was applied.

Test Results

Test	Expected Result	Actual Result
Ping before firewall	Replies received	Passed
Ping after firewall	Traffic blocked	Passed
Host protection	Unauthorized source denied	Passed

Assessment

The workstation firewall behaved as intended. Traffic from the specified remote device was denied immediately after the rule was enabled. This demonstrates that endpoint firewall policies provide an additional security layer alongside VLAN segmentation, ACLs and perimeter security devices.

Simulator Constraints

Packet Tracer does not emulate endpoint antivirus software, operating system patch management, endpoint detection and response, disk encryption, backup solutions or detailed operating system logs. These controls should be implemented in a production environment to provide complete endpoint protection.

Summary

The endpoint security test confirmed that host-based firewall rules can effectively restrict unauthorized access to individual workstations. While the simulator offers limited endpoint security capabilities, the demonstration supports the overall defense-in-depth strategy implemented throughout the network.

Edge Router Security Test Report

This report documents the security validation performed on the Edge Router in the NVIDIA Packet Tracer project. Each test verifies that the configured security controls operate as expected and help protect the router from unauthorized access and attacks.

1. Password Encryption Verification

Command Used:

show running-config

Observation:

The running configuration contains the 'service password-encryption' command and a minimum password length of eight characters. This confirms that locally configured passwords are encrypted and basic password policy is enforced.

Expected Result:

Passwords are not stored in plain text.

Result: PASS



```
Edge Router
Physical Config CLI Attributes
IOS Command Line Interface
*****
* WARNING!
* Unauthorized access to this device is strictly PROHIBITED.
* All activities are monitored and logged.
* If you are not an authorized user, DISCONNECT IMMEDIATELY.
*****

User Access Verification

Username:
Username: admin
Password:

Edge-Router#
Edge-Router#show running-config
Building configuration...

Current configuration : 2506 bytes
!
version 15.1
service timestamps log datetime msec
service timestamps debug datetime msec
service password-encryption
security passwords min-length 8
!
hostname Edge-Router
!
login block-for 60 attempts 10 within 30
!
!
```

2. Brute-Force Protection Verification

Commands Used:

show running-config | include login block-for

show login

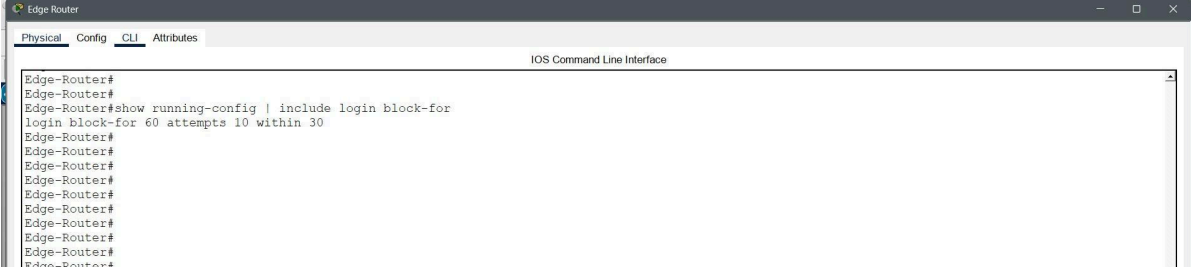
Observation:

The router is configured with 'login block-for 60 attempts 10 within 30'. The login status confirms the router is monitoring authentication failures and will temporarily block login attempts after the configured threshold.

Expected Result:

Repeated failed login attempts are blocked.

Result: PASS



```
Edge-Router#
Edge-Router#
Edge-Router#show running-config | include login block-for
login block-for 60 attempts 10 within 30
Edge-Router#
Edge-Router#
Edge-Router#
Edge-Router#
Edge-Router#
Edge-Router#
Edge-Router#
Edge-Router#
Edge-Router#
Edge-Router#
Edge-Router#
```

3. SSH Access and Session Timeout

Command Used:

show running-config | section line vty

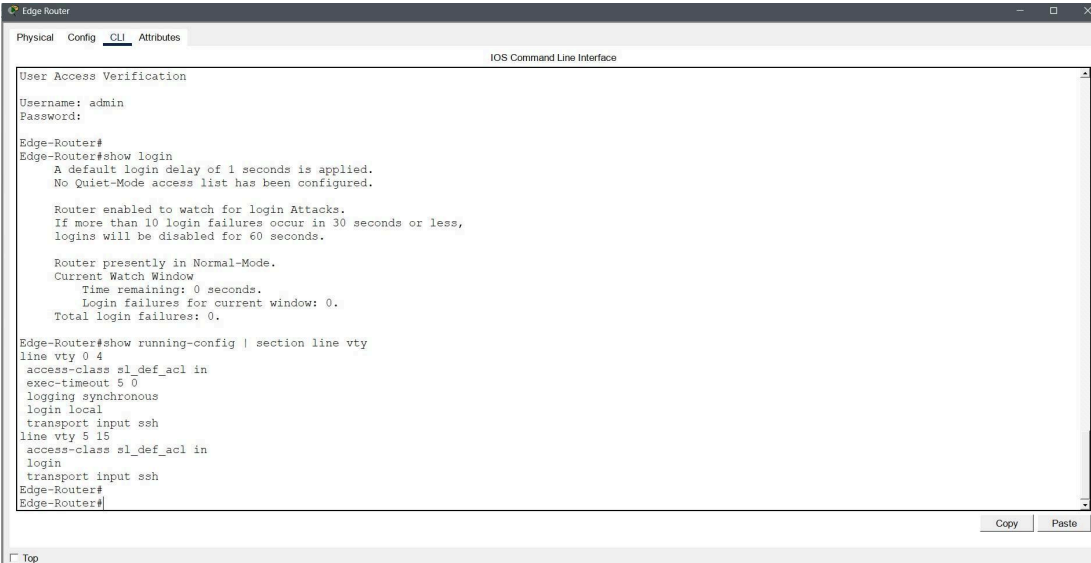
Observation:

Only SSH is allowed on the VTY lines using 'transport input ssh'. The configuration also includes 'exec-timeout 5 0', automatically disconnecting idle administrative sessions after five minutes.

Expected Result:

SSH is the only permitted remote management protocol and inactive sessions are terminated.

Result: PASS



```
User Access Verification
Username: admin
Password:
Edge-Router#
Edge-Router#show login
  A default login delay of 1 seconds is applied.
  No Quiet-Mode access list has been configured.

  Router enabled to watch for login Attacks.
  If more than 10 login failures occur in 30 seconds or less,
  logins will be disabled for 60 seconds.

  Router presently in Normal-Mode.
  Current Watch Window
  Time remaining: 0 seconds.
  Login failures for current window: 0.
  Total login failures: 0.

Edge-Router#show running-config | section line vty
line vty 0 4
 access-class a1_def_acl in
 exec-timeout 5 0
 logging synchronous
 login local
 transport input ssh
line vty 5 15
 access-class a1_def_acl in
 login
 transport input ssh
Edge-Router#
Edge-Router#
```

4. Access Control List Verification

Command Used:

show access-lists

Observation:

The management ACL blocks Telnet and permits SSH. The Internet ACL allows only approved services such as HTTP, HTTPS and DNS while denying all other inbound traffic.

Expected Result:

Unauthorized management and network traffic is blocked.

Result: PASS

```
Edge Router
Physical Config CLI Attributes
IOS Command Line Interface

Username:
Username: admin
Password:

Edge-Router#
Edge-Router#show access-lists
Extended IP access list s1_def_acl
  0 deny tcp any any eq telnet
  0 deny tcp any any eq www
  0 deny tcp any any eq 22
  0 permit tcp any any eq 22
  10 permit ip any any
Extended IP access list ACL INTERNET IN
  10 permit tcp any host 192.168.80.10 eq www
  20 permit tcp any host 192.168.80.10 eq 443
  30 permit tcp any any established
  40 permit udp any any eq domain
  50 deny ip any any
Edge-Router#
```

5. MOTD Banner Verification

Observation:

The legal warning banner is displayed before the login prompt, informing users that unauthorized access is prohibited and activity is monitored.

Expected Result:

Users receive a warning before authentication.

Result: PASS

```
Edge Router
Physical Config CLI Attributes
IOS Command Line Interface

*****
* WARNING!                                     *
* Unauthorized access to this device is strictly PROHIBITED. *
* All activities are monitored and logged. *
* If you are not an authorized user, DISCONNECT IMMEDIATELY. *
*****

User Access Verification

Username: admin
Password:

Edge-Router#
Edge-Router#show login
  A default login delay of 1 seconds is applied.
  No Quiet-Mode access list has been configured.

  Router enabled to watch for login Attacks.
  If more than 10 login failures occur in 30 seconds or less,
  logins will be disabled for 60 seconds.

  Router presently in Normal-Mode.
  Current Watch Window
    Time remaining: 0 seconds.
    Login failures for current window: 0.
    Total login failures: 0.

Edge-Router#
```

6. Syslog Logging Verification

Command Used:

show logging

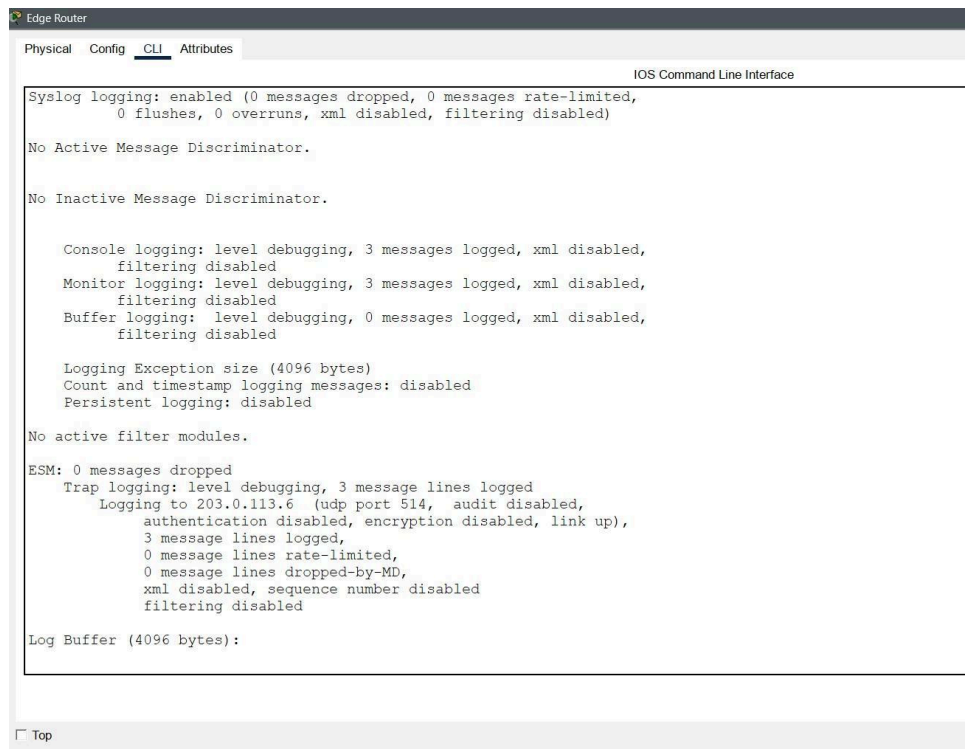
Observation:

Syslog logging is enabled and the router is forwarding logs to the remote Syslog server at 203.0.113.6 on UDP port 514. This provides centralized logging for security monitoring and auditing.

Expected Result:

Security and system events are successfully logged.

Result: PASS



```
Edge Router
Physical Config CLI Attributes
IOS Command Line Interface

Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 3 messages logged, xml disabled,
filtering disabled
Monitor logging: level debugging, 3 messages logged, xml disabled,
filtering disabled
Buffer logging: level debugging, 0 messages logged, xml disabled,
filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level debugging, 3 message lines logged
Logging to 203.0.113.6 (udp port 514, audit disabled,
authentication disabled, encryption disabled, link up),
3 message lines logged,
0 message lines rate-limited,
0 message lines dropped-by-MD,
xml disabled, sequence number disabled
filtering disabled

Log Buffer (4096 bytes):

Top
```

Conclusion

All security validation tests completed successfully. Password encryption, brute-force protection, SSH-only remote access, session timeout, management and Internet ACLs, legal login banner and centralized Syslog logging are correctly configured on the Edge Router. These controls strengthen the security posture of the router and reduce the risk of unauthorized access.

Control Plane Policing (CoPP) Configuration Test Report

This report verifies the Control Plane Policing (CoPP) configuration implemented on the Edge Router in the NVIDIA Packet Tracer project and documents the limitations of Packet Tracer for operational CoPP testing.

1. Objective

Verify that the CoPP configuration has been accepted by the router and determine whether Packet Tracer supports runtime verification.

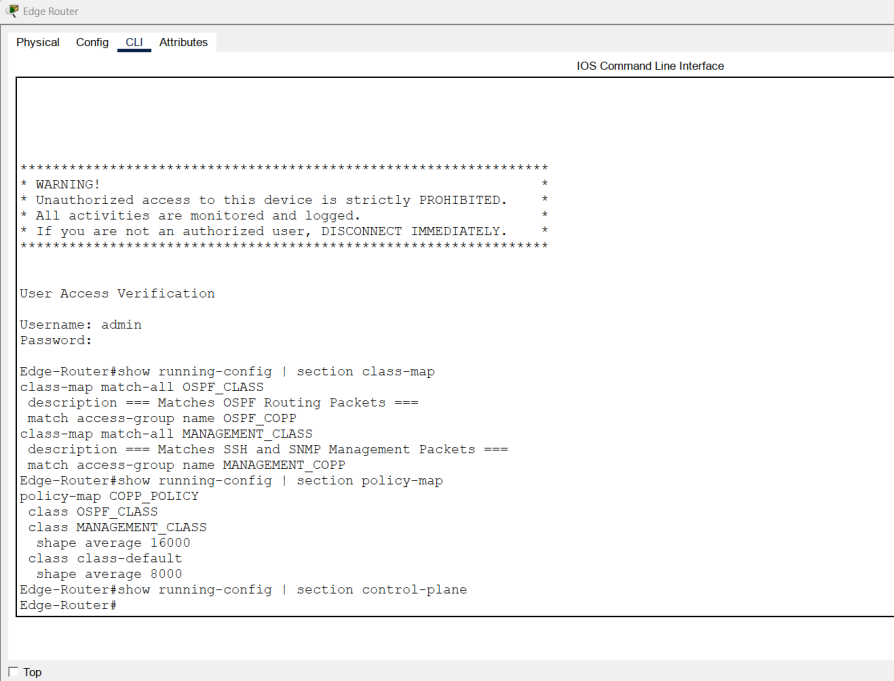
2. Configuration Verification

Commands Used:

show running-config | section class-map

show running-config | section policy-map

show running-config | section control-plane



```
Edge Router
Physical Config CLI Attributes
IOS Command Line Interface

*****
* WARNING!                                     *
* Unauthorized access to this device is strictly PROHIBITED. *
* All activities are monitored and logged. *
* If you are not an authorized user, DISCONNECT IMMEDIATELY. *
*****

User Access Verification

Username: admin
Password:

Edge-Router#show running-config | section class-map
class-map match-all OSPF_CLASS
description === Matches OSPF Routing Packets ===
match access-group name OSPF_COPP
class-map match-all MANAGEMENT_CLASS
description === Matches SSH and SNMP Management Packets ===
match access-group name MANAGEMENT_COPP
Edge-Router#show running-config | section policy-map
policy-map COPP_POLICY
class OSPF_CLASS
class MANAGEMENT_CLASS
  shape average 16000
class class-default
  shape average 8000
Edge-Router#show running-config | section control-plane
Edge-Router#
```

Observation:

The running configuration confirms that the OSPF_CLASS and MANAGEMENT_CLASS class maps and the COPP_POLICY policy map were successfully created. However, no control-plane section is present, indicating that the policy is not attached to the router control plane.

3. Operational Verification

Commands Used:

show policy-map control-plane

show policy-map interface control-plane

show control-plane

The screenshot shows the CLI of an Edge Router in Cisco Packet Tracer. The interface has tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the IOS Command Line Interface. At the top, a banner message states: "All activities are monitored and logged. * If you are not an authorized user, DISCONNECT IMMEDIATELY. *". Below this, the prompt is "Edge-Router#". The user enters "show running-config | section class-map", which returns the configuration for two class-maps: "match-all OSPF_CLASS" and "match-all MANAGEMENT_CLASS". Next, the user enters "show running-config | section policy-map", which returns the configuration for a policy-map named "COPP_POLICY" containing the two class-maps. Then, the user enters "show running-config | section control-plane", which returns an error: "% Invalid input detected at '^' marker." because the "control-plane" section is not supported in this IOS image. The user then enters "show policy-map control-plane" and "show policy-map interface control-plane", both of which also return the same error message. The prompt returns to "Edge-Router#". A "Top" link is visible at the bottom left of the CLI window.

```
Edge Router
Physical Config CLI Attributes
IOS Command Line Interface

All activities are monitored and logged.
* If you are not an authorized user, DISCONNECT IMMEDIATELY. *
*****

User Access Verification

Username: admin
Password:

Edge-Router#show running-config | section class-map
class-map match-all OSPF_CLASS
description === Matches OSPF Routing Packets ===
match access-group name OSPF_COPP
class-map match-all MANAGEMENT_CLASS
description === Matches SSH and SNMP Management Packets ===
match access-group name MANAGEMENT_COPP
Edge-Router#show running-config | section policy-map
policy-map COPP_POLICY
class OSPF_CLASS
class MANAGEMENT_CLASS
  shape average 16000
class class-default
  shape average 8000
Edge-Router#show running-config | section control-plane
Edge-Router#show policy-map control-plane
Edge-Router#show policy-map interface control-plane

% Invalid input detected at '^' marker.

Edge-Router#show control-plane
^
% Invalid input detected at '^' marker.

Edge-Router#
Top
```

Observation:

The operational verification commands returned 'Invalid input detected'. This confirms that the Cisco Packet Tracer IOS image does not support runtime CoPP monitoring or statistics.

4. Result

Configuration Verification: PASS

Operational CoPP Validation: Not Supported in Cisco Packet Tracer

5. Conclusion

The CoPP configuration syntax is valid and is successfully stored in the running configuration. However, Packet Tracer does not implement the control-plane features required to apply or verify the policy during operation. As a result, only configuration validation can be demonstrated.

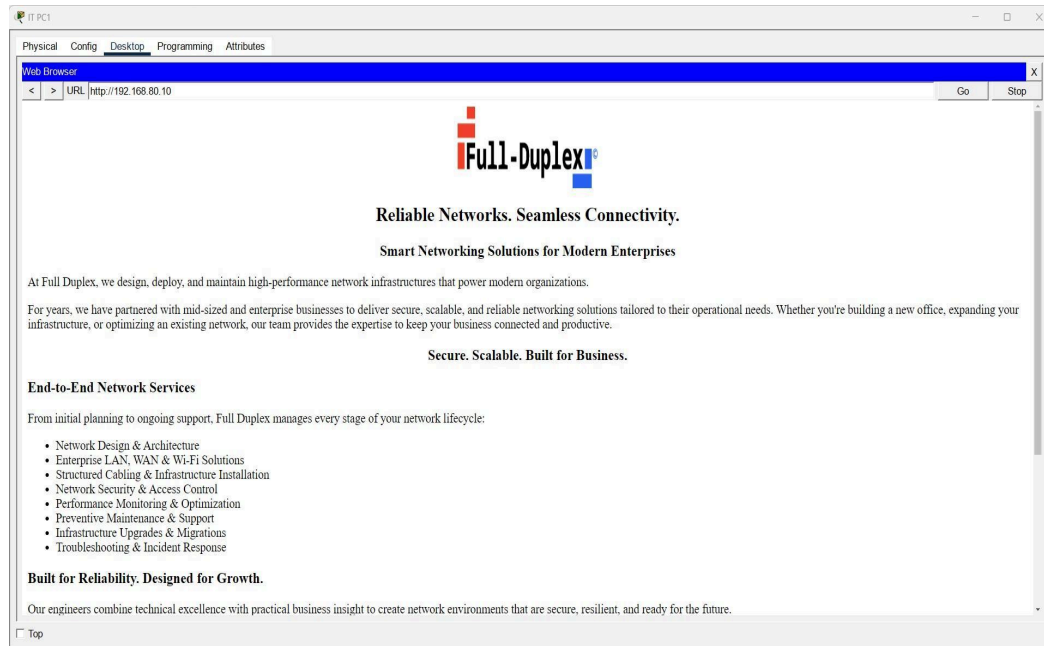
DMZ Security Validation Report

Objective

To verify that the DMZ is correctly configured to provide public services while protecting the internal network through the Cisco ASA firewall, ACLs and NAT.

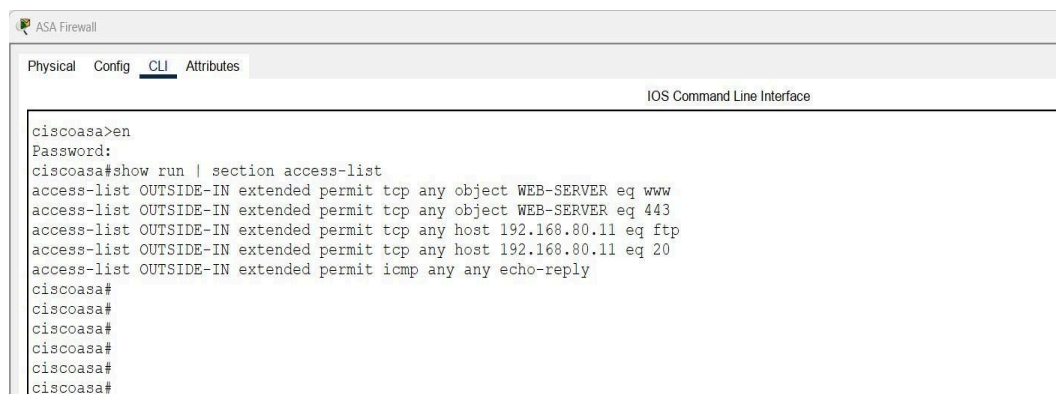
DMZ Web Access

The landing page was successfully accessed through 192.168.80.10, confirming Static NAT and HTTP connectivity.



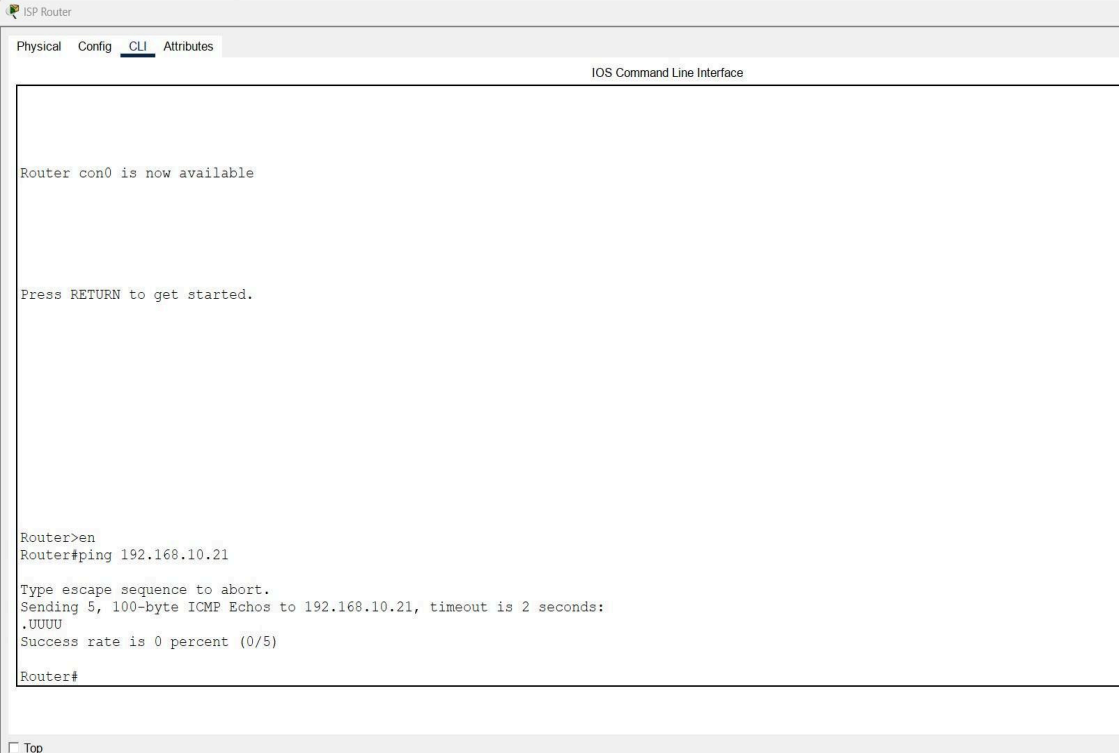
ASA ACL Verification

The ASA access list allows only required services to the DMZ while restricting unauthorized traffic.



External Access Test

An ISP router failed to reach an internal host, confirming that direct external access is blocked.



```
ISP Router
Physical Config CLI Attributes
IOS Command Line Interface

Router con0 is now available

Press RETURN to get started.

Router>en
Router#ping 192.168.10.21

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.21, timeout is 2 seconds:
.UUUU
Success rate is 0 percent (0/5)

Router#
```

Top

NAT Verification

The ASA configuration confirms Dynamic PAT for inside VLANs and Static NAT for the DMZ web server.

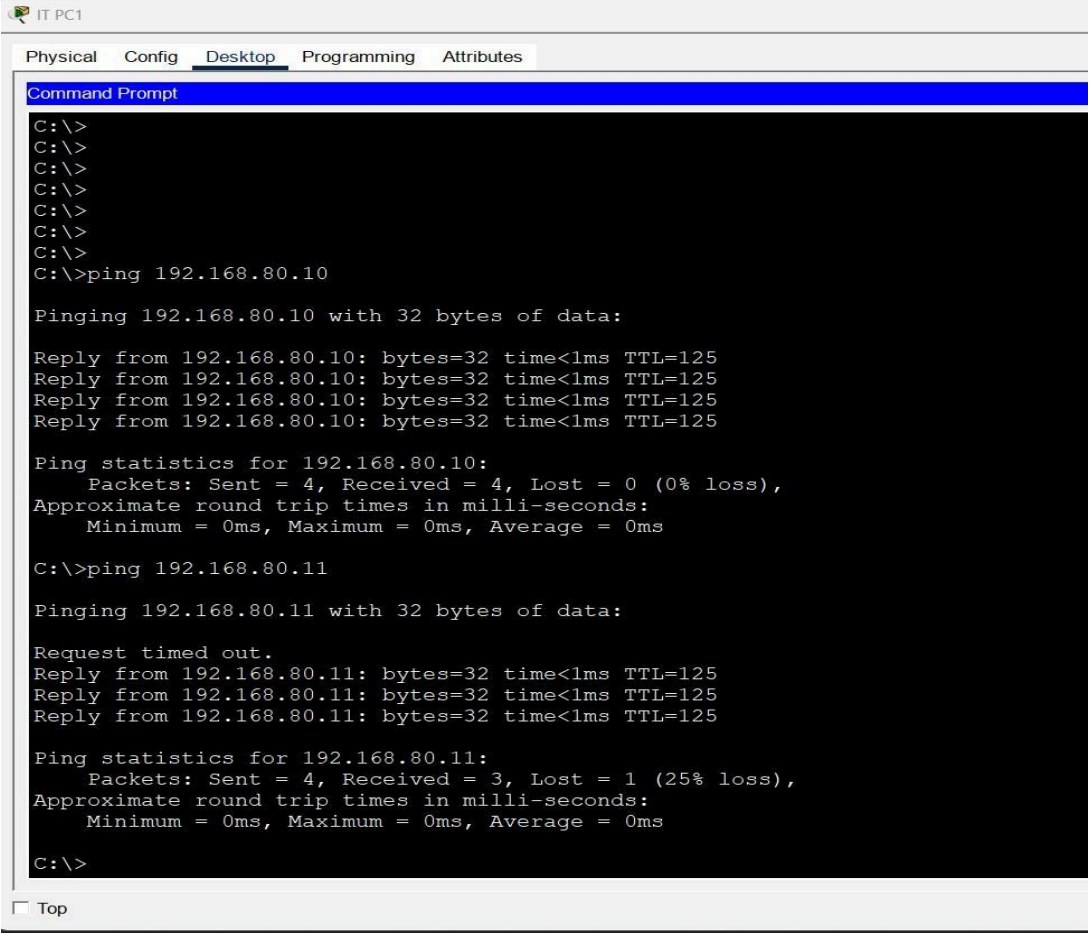


```
ASA Firewall
Physical Config CLI Attributes
IOS Command Line Interface

ciscoasa>en
Password:
ciscoasa#show running-config nat
object network INSIDE-VLANS
 nat (inside,outside) dynamic interface
object network WEB-SERVER
 nat (inside,outside) static 192.168.80.10
ciscoasa#
ciscoasa#
ciscoasa#
ciscoasa#
```

DMZ Connectivity

The internal workstation successfully pinged and accessed the DMZ web server.



```
IT PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ping 192.168.80.10

Pinging 192.168.80.10 with 32 bytes of data:

Reply from 192.168.80.10: bytes=32 time<1ms TTL=125
Reply from 192.168.80.10: bytes=32 time<1ms TTL=125
Reply from 192.168.80.10: bytes=32 time<1ms TTL=125
Reply from 192.168.80.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.80.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.80.11

Pinging 192.168.80.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.80.11: bytes=32 time<1ms TTL=125
Reply from 192.168.80.11: bytes=32 time<1ms TTL=125
Reply from 192.168.80.11: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.80.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Observation

All validation tests completed successfully. The DMZ web server is reachable through its published address while the internal network remains protected by ASA firewall policies.

Packet Tracer Limitation

Packet Tracer simulates basic ASA firewall functions such as ACLs and NAT but does not implement advanced enterprise firewall inspection and security features.

Conclusion

The DMZ configuration is operating correctly. Public services are accessible through Static NAT and firewall rules protect the internal network from unauthorized external access. Due to the limitations of Cisco Packet Tracer, an external PC could not be used to simulate Internet-based attacks. Instead, the ISP router was used to generate traffic toward the internal network. The failed ping attempts demonstrate that unauthorized access from the external network is blocked by the ASA firewall and ACL policies, confirming that the internal network is protected.

DMZ Isolation ACL Validation Report

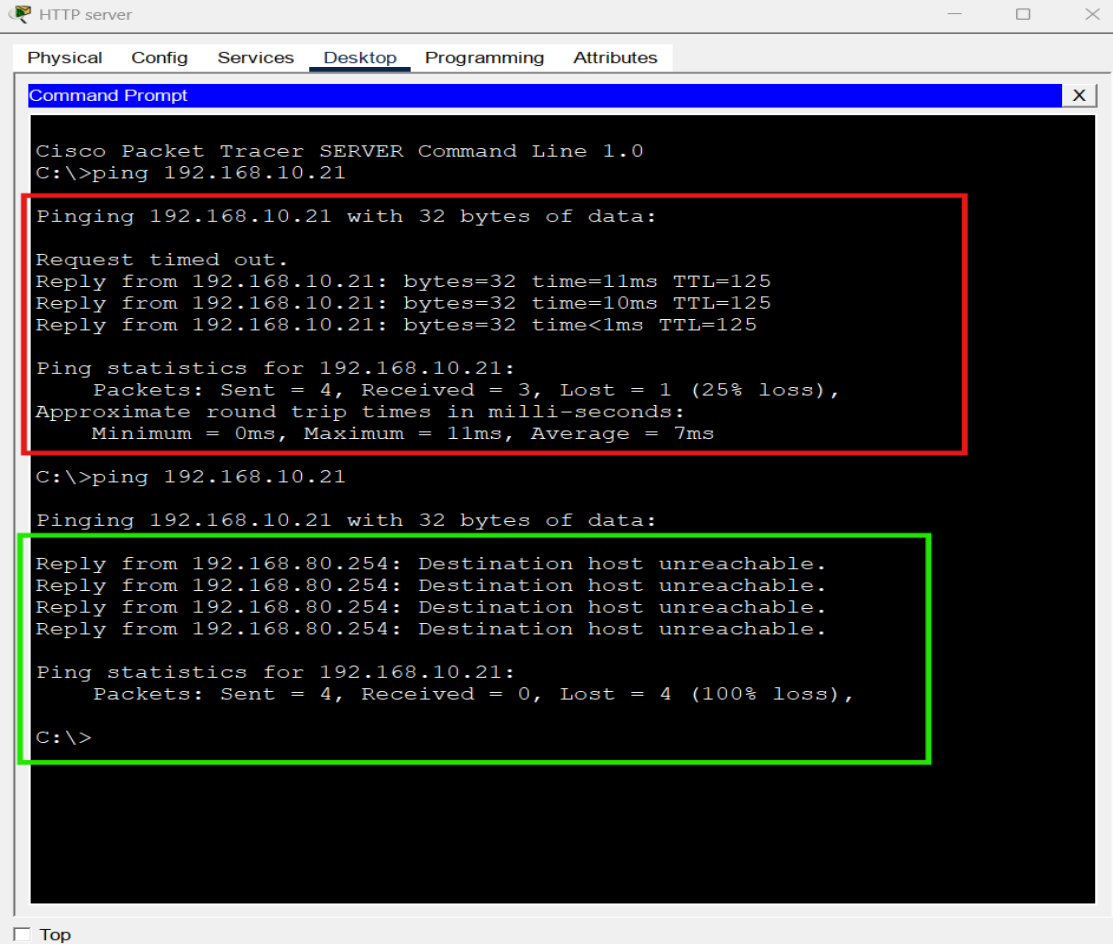
Objective

Validate that the DMZ isolation ACL prevents devices in the DMZ VLAN from communicating with the internal network after it is applied to the VLAN 80 interface.

Test Procedure

- From the DMZ HTTP server, ping the internal host (192.168.10.21) before applying the ACL.
- Apply the ACL to VLAN 80 using:
conf t
interface vlan80
ip access-group DMZ_ISOLATION_ACL in
end
- Repeat the ping test from the DMZ HTTP server.
- Compare the results.

Test Evidence



The screenshot shows a Cisco Packet Tracer window titled 'HTTP server' with tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The Command Prompt shows the output of a ping command from the DMZ HTTP server to the internal host 192.168.10.21. The first ping, before applying the ACL, shows successful results with 3 packets received and a 25% loss. The second ping, after applying the ACL, shows that all 4 packets were lost, resulting in a 100% loss.

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.10.21

Pinging 192.168.10.21 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.21: bytes=32 time=11ms TTL=125
Reply from 192.168.10.21: bytes=32 time=10ms TTL=125
Reply from 192.168.10.21: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.10.21:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 7ms

C:\>ping 192.168.10.21

Pinging 192.168.10.21 with 32 bytes of data:

Reply from 192.168.80.254: Destination host unreachable.
Reply from 192.168.80.254: Destination host unreachable.
Reply from 192.168.80.254: Destination host unreachable.
Reply from 192.168.80.254: Destination host unreachable.

Ping statistics for 192.168.10.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 1. Ping results before and after applying the DMZ isolation ACL.

Observation

Before the ACL was attached to VLAN 80 (red box), the DMZ host successfully reached the internal host after the initial ARP resolution. This indicated that the ACL was not actively filtering traffic. After the command 'ip access-group DMZ_ISOLATION_ACL in' was applied (green box), all ping requests failed with 'Destination host unreachable', showing that the Layer 3 gateway blocked traffic from the DMZ to the internal network.

Packet Tracer Limitation

During testing, Cisco Packet Tracer did not consistently preserve the interface ACL binding after the project was closed and reopened. The ACL remained in the configuration, but the interface attachment could disappear and had to be manually reapplied before testing. This behavior is a limitation of Packet Tracer rather than a fault in the network design.

Conclusion

The validation confirms that the DMZ isolation ACL functions correctly when applied to the VLAN 80 interface. The observed inconsistency is attributable to a Packet Tracer simulation limitation, not to the ACL configuration itself.

ASA Firewall Testing Report

This report summarizes the validation of the Cisco ASA firewall configuration in the NVIDIA Packet Tracer project. The objective was to verify interface status, security levels, access control lists (ACLs), NAT operation, translation table and basic connectivity.

1. Interface Verification

The ASA interfaces were checked to confirm that the inside and outside interfaces were configured correctly and were operational.

ASA Firewall

Physical Config CLI Attributes

IOS Command Line Interface

```
Type help or '?' for a list of available commands.

ciscoasa>
ciscoasa>en
Password:
ciscoasa#
ciscoasa#
ciscoasa#show ip
System IP Addresses:
Interface      Name      IP address      Subnet mask      Method
GigabitEthernet1/1  outside  203.0.113.6      255.255.255.252  manual
GigabitEthernet1/2  inside   10.0.3.2          255.255.255.252  manual
GigabitEthernet1/3              unassigned      unassigned        manual
GigabitEthernet1/4              unassigned      unassigned        unset
GigabitEthernet1/5              unassigned      unassigned        unset
GigabitEthernet1/6              unassigned      unassigned        unset
GigabitEthernet1/7              unassigned      unassigned        unset
GigabitEthernet1/8              unassigned      unassigned        unset
Management1/1              unassigned      unassigned        unset

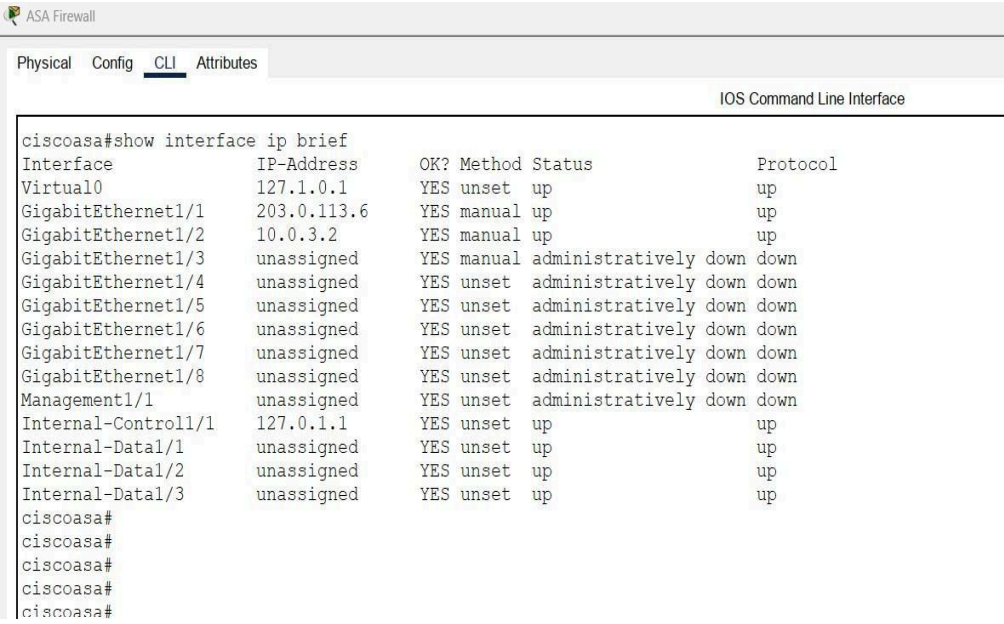
Current IP Addresses:
Interface      Name      IP address      Subnet mask      Method
GigabitEthernet1/1  outside  203.0.113.6      255.255.255.252  manual
GigabitEthernet1/2  inside   10.0.3.2          255.255.255.252  manual
GigabitEthernet1/3              unassigned      unassigned        manual
GigabitEthernet1/4              unassigned      unassigned        unset
GigabitEthernet1/5              unassigned      unassigned        unset
GigabitEthernet1/6              unassigned      unassigned        unset
GigabitEthernet1/7              unassigned      unassigned        unset
GigabitEthernet1/8              unassigned      unassigned        unset
Management1/1              unassigned      unassigned        unset

ciscoasa#show interface ip brief
```

☐ Top

2. Interface Status

The interface summary confirmed that GigabitEthernet1/1 (outside) and GigabitEthernet1/2 (inside) were up and assigned IP addresses.

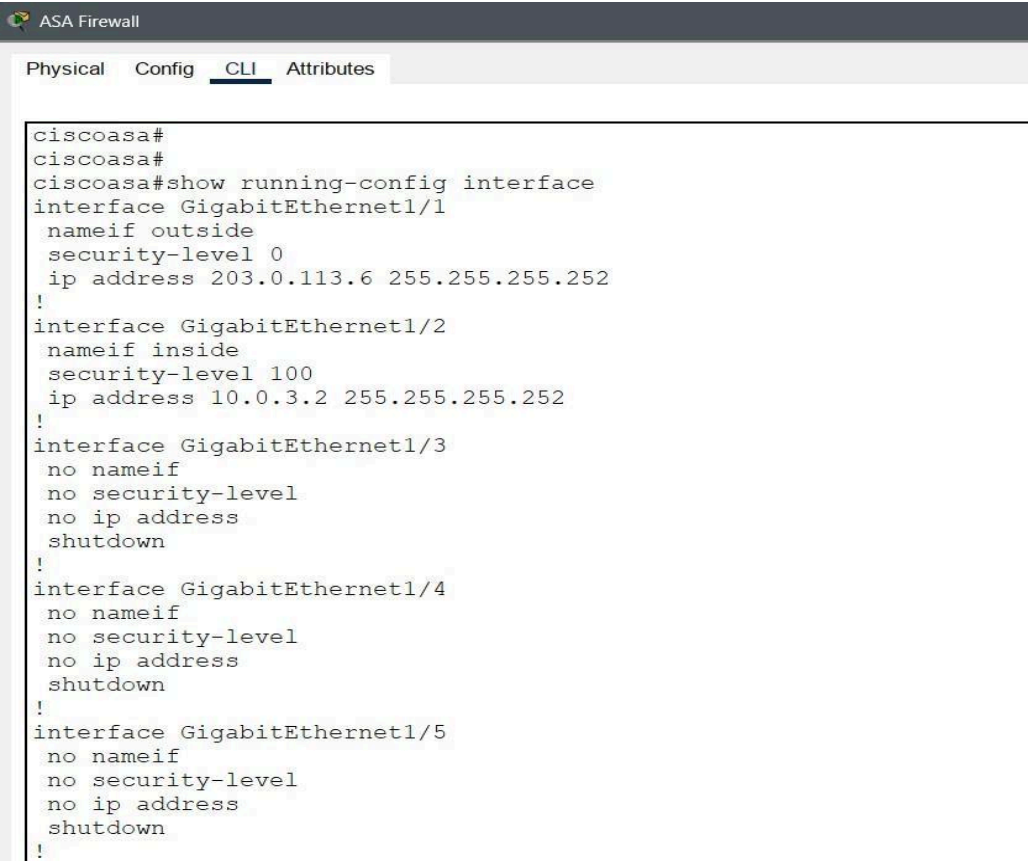


The screenshot shows the ASA Firewall CLI interface with the 'CLI' tab selected. The command 'show interface ip brief' has been executed, displaying a table of interface status. The table has columns for Interface, IP-Address, OK?, Method, Status, and Protocol. GigabitEthernet1/1 and GigabitEthernet1/2 are both 'up' with assigned IP addresses. Other interfaces are either 'administratively down' or 'unassigned'.

Interface	IP-Address	OK?	Method	Status	Protocol
Virtual0	127.1.0.1	YES	unset	up	up
GigabitEthernet1/1	203.0.113.6	YES	manual	up	up
GigabitEthernet1/2	10.0.3.2	YES	manual	up	up
GigabitEthernet1/3	unassigned	YES	manual	administratively down	down
GigabitEthernet1/4	unassigned	YES	unset	administratively down	down
GigabitEthernet1/5	unassigned	YES	unset	administratively down	down
GigabitEthernet1/6	unassigned	YES	unset	administratively down	down
GigabitEthernet1/7	unassigned	YES	unset	administratively down	down
GigabitEthernet1/8	unassigned	YES	unset	administratively down	down
Management1/1	unassigned	YES	unset	administratively down	down
Internal-Controll1/1	127.0.1.1	YES	unset	up	up
Internal-Data1/1	unassigned	YES	unset	up	up
Internal-Data1/2	unassigned	YES	unset	up	up
Internal-Data1/3	unassigned	YES	unset	up	up

3. Security Level Configuration

The running configuration confirmed that the outside interface uses security level 0 while the inside interface uses security level 100.



The screenshot shows the ASA Firewall CLI interface with the 'CLI' tab selected. The command 'show running-config interface' has been executed, displaying the configuration for interfaces GigabitEthernet1/1 through 1/5. GigabitEthernet1/1 is configured as 'outside' with security-level 0. GigabitEthernet1/2 is configured as 'inside' with security-level 100. Interfaces 1/3, 1/4, and 1/5 are configured with 'no nameif', 'no security-level', 'no ip address', and 'shutdown'.

```
ciscoasa#
ciscoasa#
ciscoasa#show running-config interface
interface GigabitEthernet1/1
  nameif outside
  security-level 0
  ip address 203.0.113.6 255.255.255.252
!
interface GigabitEthernet1/2
  nameif inside
  security-level 100
  ip address 10.0.3.2 255.255.255.252
!
interface GigabitEthernet1/3
  no nameif
  no security-level
  no ip address
  shutdown
!
interface GigabitEthernet1/4
  no nameif
  no security-level
  no ip address
  shutdown
!
interface GigabitEthernet1/5
  no nameif
  no security-level
  no ip address
  shutdown
!
```

4. Access Control List Verification

The configured ACL permits HTTP, HTTPS and FTP access to the published DMZ servers while restricting other unsolicited traffic.

```
ciscoasa#show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096) alert-interval 300
access-list OUTSIDE-IN; 5 elements; name hash: 0x58e97b67
access-list OUTSIDE-IN line 1 extended permit tcp any object WEB-SERVER eq www(hitcnt=0) 0xd1b8d607
  access-list OUTSIDE-IN line 1 extended permit tcp any host 192.168.80.10 eq www(hitcnt=0) 0x781622c8
access-list OUTSIDE-IN line 2 extended permit tcp any object WEB-SERVER eq 443(hitcnt=0) 0x1a94419a
  access-list OUTSIDE-IN line 2 extended permit tcp any host 192.168.80.10 eq 443(hitcnt=0) 0x5a1be734
access-list OUTSIDE-IN line 3 extended permit tcp any host 192.168.80.11 eq ftp(hitcnt=0) 0x0c8c8a48
access-list OUTSIDE-IN line 4 extended permit tcp any host 192.168.80.11 eq 20(hitcnt=0) 0x57cad5a
access-list OUTSIDE-IN line 5 extended permit icmp any any echo-reply(hitcnt=0) 0x07cb740e
ciscoasa#
```

5. NAT Verification

Dynamic PAT for inside users and static NAT for the web server were verified successfully.

```
ciscoasa#show nat
Auto NAT Policies (Section 2)

2 (inside) to (outside) source dynamic INSIDE-VLANS interface
  translate_hits = 4, untranslate_hits = 3
3 (inside) to (outside) source static WEB-SERVER 192.168.80.10
  translate_hits = 0, untranslate_hits = 0

ciscoasa#
```

6. Translation Table

The translation table confirmed that NAT translations were being created during traffic flow.

```
ciscoasa#show xlate
1 in use, 1 most used
Flags: D - DNS, e - extended, I - identity, i - dynamic, r - portmap, s - static, T - twice, N - net-to-net
NAT from inside:192.168.80.10/32 to outside:192.168.80.10/32 flags s idle 00:35:07, timeout 0:00:00

ciscoasa#
```

7. Connectivity Test - DMZ

The ASA successfully reached the DMZ host during testing.

```
ciscoasa#ping 192.168.80.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.80.10, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms

ciscoasa#
```

8. Connectivity Test - Outside

The ASA successfully communicated with the outside network.

```
ciscoasa#ping 200.1.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 200.1.1.1, timeout is 2 seconds:
..!!!
Success rate is 60 percent (3/5), round-trip min/avg/max = 0/0/0 ms
ciscoasa#
```

Observation

All critical firewall functions operated as expected. Interface configuration, ACLs and NAT policies were applied correctly. Connectivity tests between the ASA, DMZ and outside network were successful and active NAT translations confirmed that traffic was processed by the firewall.

Conclusion

The ASA firewall was successfully validated. The device correctly enforced security zones, applied access control policies and translated network traffic using NAT. The firewall is operating as expected for the implemented Packet Tracer design.

ACL VLAN Security Validation Report

Objective: Validate that the DMZ VLAN cannot initiate connections to the internal network while internal hosts can still access the DMZ server.

DMZ Server Configuration

The HTTP server uses 192.168.80.10/24 with gateway 192.168.80.254.

The screenshot shows the configuration page for an HTTP server. The 'IP Configuration' section is expanded, showing 'Static' IP settings. The IPv4 Address is 192.168.80.10, Subnet Mask is 255.255.255.0, and Default Gateway is 192.168.80.254. The DNS Server is 0.0.0.0. The 'IPv6 Configuration' section is also expanded, showing 'Static' settings. The IPv6 Address is FE80:2E0:A3FF:FE9D:BD49, and the Link Local Address is also FE80:2E0:A3FF:FE9D:BD49. The '802.1X' section is collapsed, showing 'Use 802.1X Security' is unchecked, and the authentication method is set to 'MD5'.

Figure: DMZ Server Configuration

ACL Configuration

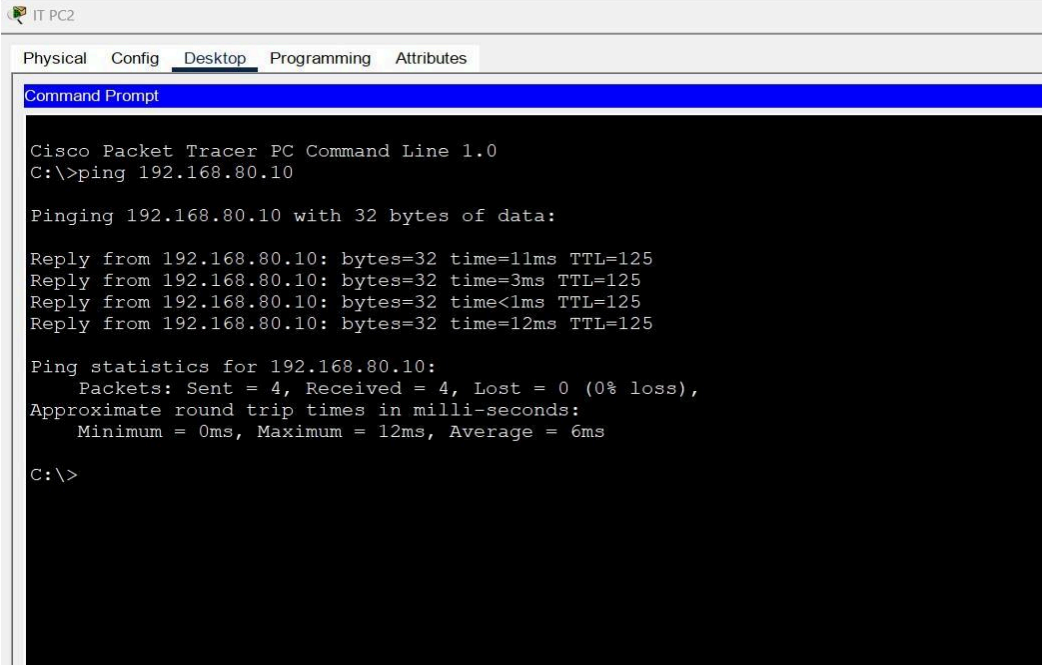
The ACL was applied using 'ip access-group DMZ_ISOLATION_ACL in'. Packet Tracer accepted the command.

The screenshot shows the CLI interface of a Leaf-3 switch. The user 'admin' has logged in successfully. The configuration commands entered are: 'interface vlan 80', 'ip access-group DMZ_ISOLATION_ACL in', and 'end'. The switch has accepted the configuration, and the user is returned to the prompt. The output shows the configuration of the DMZ_ISOLATION_ACL on interface vlan 80.

Figure: ACL Configuration

Internal to DMZ Test

The internal PC successfully reached the DMZ server.



```
IT PC2
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.80.10

Pinging 192.168.80.10 with 32 bytes of data:

Reply from 192.168.80.10: bytes=32 time=11ms TTL=125
Reply from 192.168.80.10: bytes=32 time=3ms TTL=125
Reply from 192.168.80.10: bytes=32 time<1ms TTL=125
Reply from 192.168.80.10: bytes=32 time=12ms TTL=125

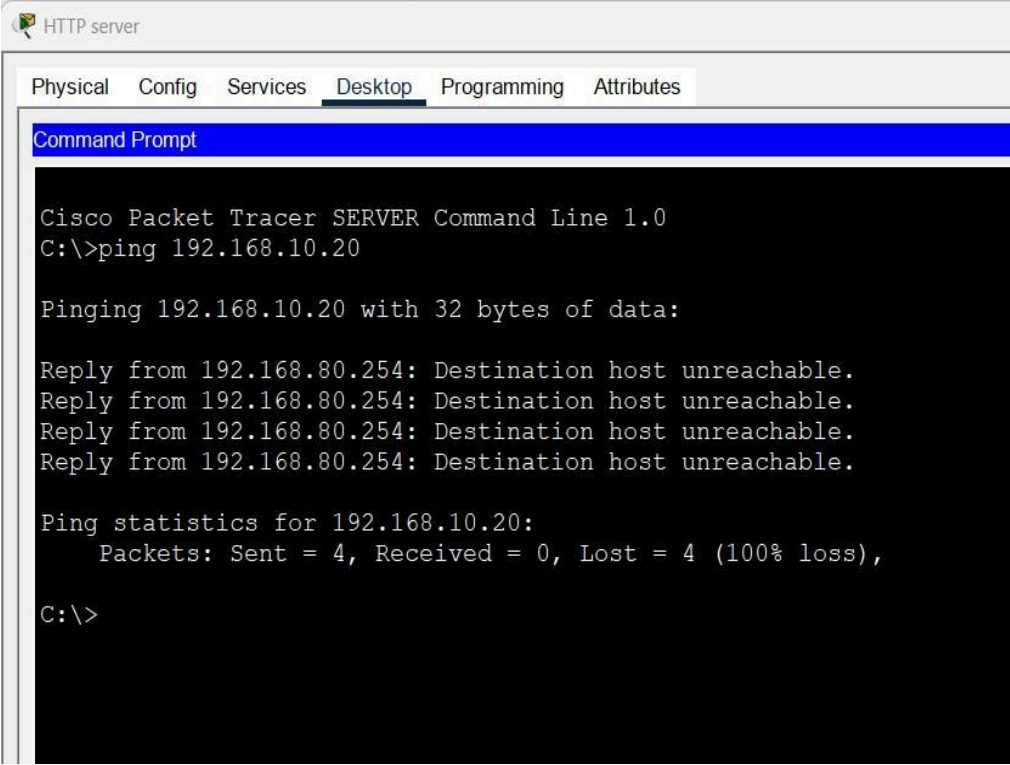
Ping statistics for 192.168.80.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 6ms

C:\>
```

Figure: Internal to DMZ Test

DMZ to Internal Test

The DMZ server received 'Destination host unreachable' when pingging the internal host.



```
HTTP server
Physical Config Services Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.80.254: Destination host unreachable.
Reply from 192.168.80.254: Destination host unreachable.
Reply from 192.168.80.254: Destination host unreachable.
Reply from 192.168.80.254: Destination host unreachable.

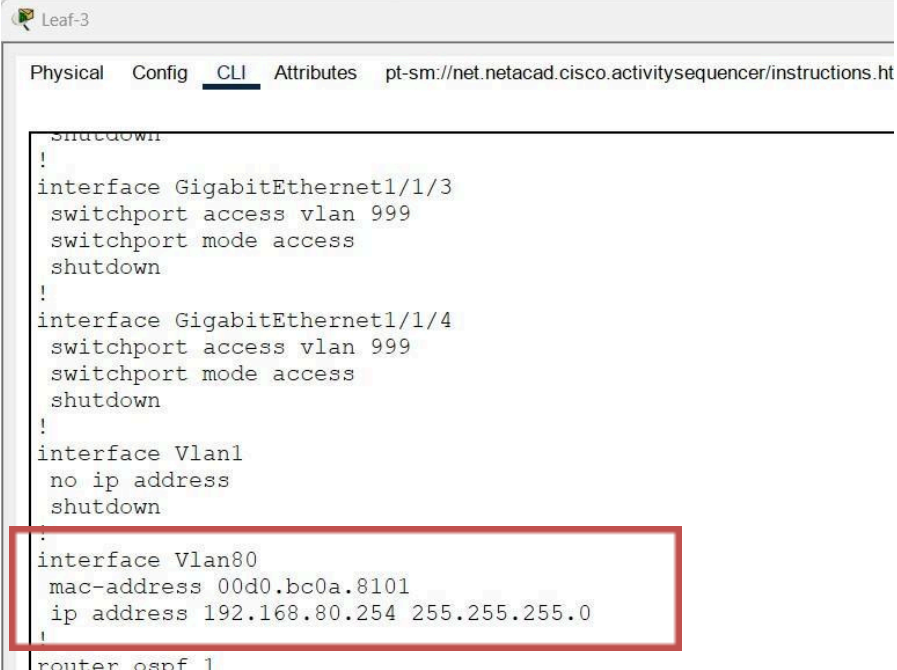
Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure: DMZ to Internal Test

Packet Tracer Limitation

The ACL binding is not shown under the VLAN interface despite the command being accepted.



```
Leaf-3
Physical Config CLI Attributes pt-sm://net.netacad.cisco.activitysequencer/instructions.ht
!
interface GigabitEthernet1/1/3
 switchport access vlan 999
 switchport mode access
 shutdown
!
interface GigabitEthernet1/1/4
 switchport access vlan 999
 switchport mode access
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan80
 mac-address 00d0.bc0a.8101
 ip address 192.168.80.254 255.255.255.0
!
router ospf 1
```

Figure: Packet Tracer Limitation

Observation

Testing shows that internal devices can access the DMZ while the DMZ cannot initiate connections to the internal network. The gateway blocks the traffic.

Problem

Packet Tracer accepts the ACL configuration command but does not display the ACL under 'show ip interface vlan 80' or in the interface running configuration. This is a known simulation limitation and can lead to confusion during validation.

Conclusion

The traffic behavior confirms that the intended DMZ isolation policy is working. The missing interface ACL display should be documented as a Packet Tracer limitation.

Thank You

Thank you for taking the time to review this project report.

Your time, feedback and consideration are sincerely appreciated. This project represents the effort invested in designing, implementing and validating a secure enterprise network using Cisco Packet Tracer. I hope this report clearly demonstrates the technical concepts, security controls and testing performed throughout the project.

Disclaimer

This project was developed and tested using Cisco Packet Tracer. Due to the limitations of the simulator, some enterprise networking features cannot be fully implemented or validated. The results presented in this report demonstrate the intended network design and security concepts within the capabilities of the simulation environment.